

Supplementary Materials for
Title name

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1 Preliminary on proximal standardization formulas

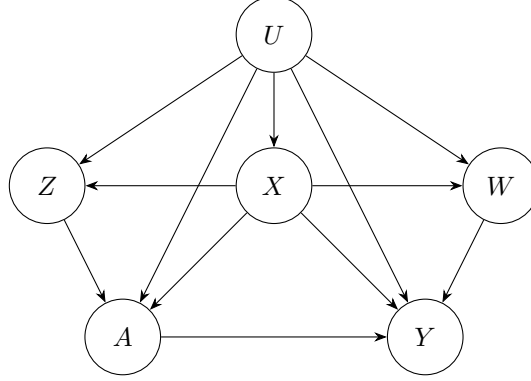


Figure 1: A DAG for proximal counterfactual model

Standardization in proximal causal framework is closely related to the proxy variables by the existence of the unobservable confounders. There are two classes of proxies. For those connecting both unobservable confounders and outcome, we call them outcome confounding proxies represented by W in Figure 1, while variables lying in between the treatment and unobservable confounders are treatment confounding proxies which is the node Z . Miao et al.[7] put forward a standardization formula using the outcome confounding proxies and Cui et al.[16] managed to identify the ATE with the treatment confounding proxies. Their identification methods require solving functions from Fredholm integral equations of the first kind. The solutions are called bridge functions because they remove the influence from the unobservable confounders and make it possible to identify ATE only with the observed. Moreover, the standardization formulas also depend on the completeness assumption on the distribution families of the confounder variables. We restate the important assumptions proposed by the pioneers in proximal causal inference as baseline for the later derivations.

Assumption 1.1 (*Outcome confounding standardization formula*[7])

1. **Completeness:** The distribution family for confounder variables U, Z, A, X satisfies that for any square-integrable function g and for any a, x , $E[g(U)|Z, A = a, X = x] = 0$ almost surely if and only if $g(U) = 0$ almost surely.
2. **Existence of outcome confounding bridge function:** Denote the singular system of the conditional expectation operator $T : L_2(P_{(W,A,X)}) \mapsto L_2(P_{(Z,A,X)})$ by $(\sigma_i, u_i, v_i)_{i \geq 1}$. Then Picard's condition $\sum_{i \geq 1} \frac{|E[Y|Z, A=a, X, v_i]|^2}{\sigma_i^2} < \infty$ holds to make integral equation

$$E[Y|Z, A = a, X] = \int h_a(w, A, X) dF(w|Z, A = a, X) = E[h_a(W, a, X)|Z, A = a, X], \quad (1)$$

solvable.

Assumption 1.2 (*Treatment confounding standardization formula*[16])

1. **Completeness:** The distribution family for confounder variables U, W, A, X satisfies that for any square-integrable function g and for any a, x , $E[g(U)|W, A = a, X = x] = 0$ almost surely if and only if $g(U) = 0$ almost surely.
2. **Existence of treatment confounding bridge function:** Denote the singular system of the conditional expectation operator $T^* : L_2(P_{(Z,A,X)}) \mapsto L_2(P_{(W,A,X)})$ by $(\sigma_i^*, u_i^*, v_i^*)_{i \geq 1}$. Then Picard's condition $\sum_{i \geq 1} \frac{|\langle \frac{1}{f(a|W, X)}, v_i^* \rangle|^2}{(\sigma_i^*)^2} < \infty$ holds to make integral equation solvable

$$\frac{1}{f(a|W, X)} = \int q_a(z, A, X) dF(z|W, A = a, X) = E[q_a(Z, a, X)|W, A = a, X], \quad (2)$$

where $f(a|W, X) = Pr\{A = a|W, X\}$.

If Assumptions 1.1 and 1.2 hold, then we have the two standardization formulas,

$$\begin{aligned} E[Y^a] &= E[h_a(W, a, X)], \\ E[Y^a] &= E[Y q_a(Z, a, X) \mathbb{1}_{A=a}]. \end{aligned}$$

for all categories of a .

2 Kernel embedded doubly robust ATE estimator

To estimate ATE in proximal causal framework, we adopt the dual kernel embedding method which was proposed by Dai et al.[4] to address the problem of learning from conditional distributions. These problems can be viewed as solving for functions that link conditional distributions to target values. Through a series of reformulations, including ERM reformulation, minimax problem reformulation through Fenchel duality and interchange of minimization and integration, and finally the kernel embeddings of means and cross-covariances, the method transforms the conditional expectation equation into a minimax problem for kernel embeddings of mean and (cross-)covariance. The method is based on the definition of the universal kernel[13], whereby any bounded continuous function can be well approximated by functions belonging to the reproducing kernel Hilbert space (RKHS) induced by a universal kernel. Consequently, the solution obtained via kernel embedding can approximate the original bounded continuous function arbitrarily well. Dai et al. proposed multiple stochastic gradient descent based algorithms to calculate functions of interest numerically and discussed the convergence behavior.

Following Dai et al., Muandet et al.[8] applied the dual kernel embedding method to instrumental variable problems and obtained a closed-form solution in RKHS to approximate the causal functions in instrumental variable regression. Given the similarity between the integral equation for establishing the existence of bridge functions and instrumental variable regression, we apply the dual kernel embedding method to approximate the true boundedly continuous bridge functions q_a and h_a on RKHSs. And then to estimate ATE by plugging estimated bridge functions in the empirical form of the doubly robust standardization formula[16]

$$\chi = E[\mathbb{1}_{A=1} q_1(Z, 1, X)[Y - h_1(W, 1, X)] - \mathbb{1}_{A=0} q_0(Z, 0, X)[Y - h_0(W, 0, X)] + h_1(W, 1, X) - h_0(W, 0, X)]. \quad (3)$$

We start by giving a detailed derivation of finding the closed-form kernel embedded approximator q_a^* on a RKHS of q_a and its empirical form $\hat{q}_a^*$. Then derive h_a^* and $\hat{h}_a^*$ with the simplest description.

2.1 Tikhonov regularized kernel embedded treatment confounding bridge function

The dual kernel embedding method relies on the original integral equation (2), however which contains unknown propensity score $f(a|W, X)$. We give the equivalent form of this hard-to-tackle problem by tower property of conditional expectation.

$$\begin{aligned} \frac{1}{f(a|W, X)} &= E[q_a(Z, a, X)|W, A = a, X] \\ 1 &= E[q_a(Z, a, X)|W, A = a, X] f(a|W, X) \\ 1 &= E_A[E_Z[\mathbb{1}_{A=a} q_a(Z, a, X)|W, A, X]|W, X] \\ 1 &= E_{ZA}[\mathbb{1}_{A=a} q_a(Z, a, X)|W, X]. \end{aligned} \quad (4)$$

Then we perform the ERM reformulation of (4).

Lemma 2.1 (*ERM reformulation*)

The true solution $q_a^*(Z, a, X)$ for integral equation (4) solves the expected risk minimization problem with square loss for 1-dim data $l(x, y) = \frac{1}{2}(x - y)^2$

$$\min_{q_a \in L_2(P_{(Z, X|W, X)})} E_{W, X}[l(1, E_{ZA}[\mathbb{1}_{A=a} q_a(Z, a, X)|W, X])]. \quad (5)$$

Proof:

First we consider the solution of $E_{W,X}[\frac{1}{2}(1-S(W,X))^2]$. For all $S \in L_2(P_{(W,X)})$, since $E_{W,X}[\frac{1}{2}(1-S(W,X))^2] \geq 0$ and it reaches the minimum 0 within a function class $\{S : E[S(W,X)] = 1\} \subseteq L_2(P_{(W,X)})$, it is clear that $E_Z[\mathbb{1}_{A=a}q_a^*(Z,a,X)|W,X] \in \{S : E[S(W,X)] = 1\}$ by Equation (4). Hence the true solution $q_a^*(Z,a,X)$ for integral equation (4) solves the ERM problem. $\square$

Next comes the minimax problem reformulation of ERM form.

Lemma 2.2 (Minimax problem reformulation)

The ERM form (5) of integral equation (4) is equivalent to a minimax problem

$$\min_{q_a \in L_2(P_{(Z,X|W,X)})} \max_{u(\cdot, \cdot) \in \mathcal{M}(W,X)} E_{Z,W,A,X}[(\mathbb{1}_{A=a}q_a(Z,a,X) - 1)u(W,X)] - \frac{1}{2}E_{W,X}[u^2(W,X)], \quad (6)$$

where $\mathcal{M}(W,X)$ is the space of all measurable functions from $\mathcal{W} \times \mathcal{X} \mapsto \mathbb{R}$.

Proof:

For simplicity, we denote the square loss function $l(x,y)$ by $l_x(y)$. Its convex conjugate function is defined as $l_x^*(y) := \max_{u \in \mathbb{R}}\{uy - l_x(u)\}$.

$$\begin{aligned} & \min_{q_a \in L_2(P_{(Z,X|W,X)})} E_{W,X}[l_1(E_{ZA}[\mathbb{1}_{A=a}q_a(Z,a,X)|W,X])] \\ &= \min_{q_a \in L_2(P_{(Z,X|W,X)})} E_{W,X}[\max_{u \in \mathbb{R}}\{uE_{ZA}[\mathbb{1}_{A=a}q_a(Z,a,X)|W,X] - l_1^*(u)\}] \\ &= \min_{q_a \in L_2(P_{(Z,X|W,X)})} E_{W,X}[\max_{u \in \mathbb{R}}\left\{uE_{ZA}[\mathbb{1}_{A=a}q_a(Z,a,X)|W,X] - u - \frac{1}{2}u^2\right\}] \end{aligned} \quad (7)$$

$$\begin{aligned} &= \min_{q_a \in L_2(P_{(Z,X|W,X)})} \max_{u(\cdot, \cdot) \in \mathcal{M}(W,X)} E_{W,X}[u(W,X)E_{ZA}[\mathbb{1}_{A=a}q_a(Z,a,X)|W,X] - u(W,X) - \frac{1}{2}u^2(W,X)] \quad (8) \\ &= \min_{q_a \in L_2(P_{(Z,X|W,X)})} \max_{u(\cdot, \cdot) \in \mathcal{M}(W,X)} E_{Z,W,A,X}[(\mathbb{1}_{A=a}q_a(Z,a,X) - 1)u(W,X)] - \frac{1}{2}E_{W,X}[u^2(W,X)]. \end{aligned}$$

Since the square loss function $\frac{1}{2}(x-y)^2$ is a convex and continuous real function on $\mathbb{R}$ given a fixed x , (l_1^*, l_1) are dual to each other, which forms a Fenchel duality[10]. Therefore, we have $l_x(y) := \max_{u \in \mathbb{R}}\{uy - l_x^*(u)\}$. Substituting l_1^* by the explicit form $l_1^*(u) = u + \frac{1}{2}u^2$, we get the equivalent expression (7).

The transformation from (7) to (8) is an interchange of maximization and integration. It relies on the property that l_1^* is a normal integrand and that the space of all measurable functions from $\mathcal{W} \times \mathcal{X}$ to $\mathbb{R}$ is decomposable. One can check definition 14.27 and definition 14.59 about normal integrands and decomposable spaces, and theorem 14.60 on the interchange of minimization and integration of [10] for detailed introduction. $\square$

To remain consistent with the terminology used by Dai et al. [4], we will continue to refer to $u(W,X)$ as the dual function. Below we give two important properties of the optimal dual function $u^*(W,X)$.

Proposition 2.1 (Uniqueness)

The optimal dual function u^* is uniquely defined by $u^*(W,X) = E_{ZA}[\mathbb{1}_{A=a}q_a(Z,a,X)|W,X] - 1$.

The uniqueness of u^* is due to the strict concavity of the function in minimax problem (6). Its explicit form is derived by the derivative with respect to u .

Proposition 2.2 (Bounded continuity)

The optimal dual function $u^*(W,X)$ is bounded continuous on $\mathcal{W} \times \mathcal{X}$ if

- for $a \in \mathcal{A}$, $\mathbb{1}_{A=a}q_a(Z, a, X)$ is continuous in both Z and X or at least continuous in Z for any X ;
- the conditional density $p(Z|W, a, X)$ and conditional probability $f(a|W, X)$ are continuous at any $(W, X) \in \mathcal{W} \times \mathcal{X}$.

Proof:

The two continuity assumptions make sure $E_{ZA}[\mathbb{1}_{A=a}q_a(Z, a, X)|W, X]$ is a bounded continuous function of (W, X) . Since the derivative of square loss $l_1(\cdot)$ is continuous, combining the result in Proposition 2.1, we have the bounded continuity of the optimal dual function. $\square$

Proposition 2.1 and 2.2 show that the optimal dual function $u^*(W, X)$ is uniquely defined on the space of all bounded continuous functions $\mathcal{W} \times \mathcal{X} \mapsto \mathbb{R}$, which we denote by $\mathcal{C}_0(W, X)$.

To commence further transforms about the minimax problem (6), we need the following assumption.

Assumption 2.1 (*Continuous treatment confounding bridge function*)

The Picard's condition for the existence of the treatment confounding bridge function holds in a way such that there exists at least one bounded continuous treatment confounding bridge function for each treatment.

Since $\mathcal{C}_0(W, X) \subseteq \mathcal{M}(W, X)$ and $\mathcal{C}_0(Z, X) \subseteq L_2(P_{(Z, X|W, X)})$ over any polish subspaces of $\mathbb{R}$, under Assumption 2.1, we can restrict the minimax problem (6) to

$$\min_{q_a \in \mathcal{C}_0(Z, X)} \max_{u \in \mathcal{C}_0(W, X)} E_{Z, W, A, X}[(\mathbb{1}_{A=a}q_a(Z, a, X) - 1)u(W, X)] - \frac{1}{2}E_{W, X}[u^2(W, X)]. \quad (9)$$

By the definition of universal kernels, the RKHS induced by a universal kernel is dense in the space of bounded continuous functions. We can further restrict (9) to a minimax problem finding solutions in RKHSs. This gives

$$\min_{q_a \in F} \max_{u \in H} E_{Z, W, A, X}[(\mathbb{1}_{A=a}q_a(Z, a, X) - 1)u(W, X)] - \frac{1}{2}E_{W, X}[u^2(W, X)]. \quad (10)$$

Although we have changed the searching space of optimal functions to RKHSs, it is still untractable to solve (10) when there are conditional expectations. Based on this consideration, Smola et al.[11] and Song et al.[12] discussed kernel embeddings of mean and (cross-)covariance. They are used to compute the expectations and covariances of RKHS functions without distributions of random variables but only with inner product between RKHS elements. This helps transfer (10) to a kernel embedded minimax problem.

Lemma 2.3 (*Kernel embedded reformulation*)

For any two universal kernels $k: (\mathcal{W} \times \mathcal{X}) \times (\mathcal{W} \times \mathcal{X}) \mapsto \mathbb{R}$ and $\tilde{l}: (\mathcal{Z} \times \mathcal{X}) \times (\mathcal{Z} \times \mathcal{X}) \mapsto \mathbb{R}$, we denote H , $\tilde{F}$ and F by RKHSs induces by k , $\tilde{l}$ and l , where $l: (\mathcal{Z} \times \mathcal{A} \times \mathcal{X}) \times (\mathcal{Z} \times \mathcal{A} \times \mathcal{X}) \mapsto \mathbb{R}$ is defined by $l((Z, A, X), (Z', A', X')) := \mathbb{1}_{A=A'}\tilde{l}((Z, X), (Z', X'))$. The corresponding feature maps of k , $\tilde{l}$ and l are given by $\phi: \mathcal{W} \times \mathcal{X} \mapsto H$, $\tilde{\psi}: \mathcal{Z} \times \mathcal{X} \mapsto \tilde{F}$ and $\psi: \mathcal{Z} \times \mathcal{A} \times \mathcal{X} \mapsto F$. Denote the Hilbert-Schmidt Riesz representatives of (cross-)covariance operators by the operator $C_{(ZAX)(WX)}$ from F to H and the operator $C_{(WX)}$ from H to H respectively. Denote the mean embedding on H by $\mu_{(WX)}$. The kernel embedded form of (10) is given by

$$\min_{q_a \in F} \max_{u \in H} \Gamma(q_a, u), \text{ for } \Gamma(q_a, u) := \left\langle C_{(ZAX)(WX)}q_a - \mu_{(WX)} - \frac{1}{2}C_{(WX)}u, u \right\rangle_H, \quad (11)$$

where the operators $C_{(ZAX)(WX)}$ and $C_{(WX)}$ are not centered.

Proof:

First notice that $F = \tilde{F} \oplus \tilde{F}$ and $q_a \in \tilde{F} \cap F$. Since by Moore-Aronszajn theorem [5], q_a has a representation in F which is given by

$$q_a = \sum_{i: a_i=a} \alpha_i \tilde{\psi}(z_i, x_i) = \sum_{i \geq 1} \alpha_i \mathbb{1}_{a_i=a} \tilde{\psi}(z_i, x_i) = \sum_{i \geq 1} \alpha_i \psi(z_i, a_i, x_i).$$

This gives

$$\begin{aligned}
E_{WX}[u(W, X)] &= \langle \mu_{(WX)}, u \rangle_H, \\
E_{Z, W, A, X}[\mathbb{1}_{A=a} q_a(Z, a, X) u(W, X)] &= E_{Z, W, A, X}[\mathbb{1}_{A=a} \langle q_a, \tilde{\psi}(Z, X) \rangle_{\tilde{F}} \langle u, \phi(W, X) \rangle_H] \\
&= E_{Z, W, A, X} \left[\langle q_a, \underbrace{\mathbb{1}_{A=a} \tilde{\psi}(Z, X)}_{\psi(Z, A, X)} \rangle_F \langle u, \phi(W, X) \rangle_H \right] \\
&= \langle q_a \otimes u, E_{Z, W, A, X}[\psi(Z, A, X) \otimes \phi(W, X)] \rangle_{HS(H, F)} \\
&= \langle q_a \otimes u, C_{(WX)(ZAX)} \rangle_{HS(H, F)} \\
&= \langle q_a, C_{(WX)(ZAX)} u \rangle_F \\
&= \langle C_{(ZAX)(WX)} q_a, u \rangle_H.
\end{aligned}$$

Hence the kernel embedded form of (10) is given by (11). □

Notice that although the kernel embedded form (11) is concave relative to u and convex about q_a , the uniqueness of the saddle point does not hold since the convexity and concavity are not strict. If the saddle point (q_a^*, u^*) exists, then by first order condition, it must satisfy

$$\begin{aligned}
\nabla_{q_a} \Gamma \Big|_{(q_a=q_a^*, u)} &= C_{(WX)(ZAX)} u = 0 \\
\nabla_u \Gamma \Big|_{(q_a, u=u^*)} &= C_{(ZAX)(WX)} q_a - \mu_{(WX)} - C_{(WX)} u^* = 0.
\end{aligned}$$

Hence the sufficient condition for the existence of saddle points is that

1. $u \in \text{Null}(C_{(WX)(ZAX)})$;
2. $C_{(WX)} u^* + \mu_{(WX)} \in \text{Range}(C_{(ZAX)(WX)})$.

Furthermore, to determine the uniqueness of the saddle point, it must be assumed that operators

$$C_{WX} : H \mapsto H \text{ and } C_{(WX)(ZAX)} C_{(WX)}^{-1} C_{(ZAX)(WX)} : F \mapsto F$$

are invertible. However, by the Riesz-Schauder theorem [9], if the Hilbert space is infinite dimensional, then 0 belongs to the spectrum of the compact operator. This means the compact operator like C_{WX} acting on an infinite dimensional Hilbert space can't be boundedly invertible. Hence the invertibility assumption restricts H and F to be finite dimensional, which isn't true in general for RKHSs induced by universal kernels. So, the series of transforms from the original inverse problem (4) to kernel embedded minimax problem (11) have not eliminated the ill-posedness of the original problem but shifted it to an ill-posed problem that can be handled by regularization.

Hence, we seek to find the solution of the regularized form of kernel embedded minimax problem. It is given by

$$\min_{q_a \in F} \max_{u \in H} \Gamma_{\lambda_1, \lambda_2}(q_a, u), \text{ for } \Gamma_{\lambda_1, \lambda_2}(q_a, u) := \left\langle C_{(ZAX)(WX)} q_a - \mu_{(WX)} - \frac{1}{2} C_{(WX)} u, u \right\rangle_H - \frac{1}{2} \lambda_1 \|u\|_H^2 + \frac{1}{2} \lambda_2 \|q_a\|_F^2, \quad (12)$$

where λ_1 and λ_2 guarantee the bounded invertibility of $C_{(WX)} + \lambda_1 I$ and $C_{(WX)(ZAX)}(C_{(WX)} + \lambda_1 I)^{-1} C_{(ZAX)(WX)} + \lambda_2 I$ with the identity operator I .

The regularization terms make the kernel embedded minimax problem strictly convex in q_a and strictly concave in u . This means there exists a unique saddle point with a closed-form expression.

Theorem 2.1 (*Regularized kernel embedded solutions*)

The regularized kernel embedded minimax problem (12) has a unique saddle point (q_a^*, u^*) given by

$$\begin{aligned}
u^* &= (C_{(WX)} + \lambda_1 I)^{-1} (C_{(ZAX)(WX)} q_a - \mu_{(WX)}), \\
q_a^* &= (C_{(WX)(ZAX)} (C_{(WX)} + \lambda_1 I)^{-1} C_{(ZAX)(WX)} + \lambda_2 I)^{-1} C_{(WX)(ZAX)} (C_{(WX)} + \lambda_1 I)^{-1} \mu_{(WX)}.
\end{aligned}$$

Proof:

By Example 3.4.1 of [1], the saddle point of a minimax problem is unique only if the second derivatives on the saddle point is negative definite relative to the maximizer and positive definite about the minimizer. In (12), the uniqueness of the saddle point derives from the strict concavity relative to u and strict convexity relative to q_a , which means the negative definiteness and positive definiteness of second Fréchet derivative of u and q_a .

The unique optimal kernel embedded dual function $u^* \in H$ satisfies the first order condition of the extreme value point. Setting Fréchet derivative relative to u about $\Gamma_{\lambda_1, \lambda_2}(q_a, u)$ at $(q_a, u = u^*)$ to zero, by the self-adjointness of $C_{(WX)}$, we have

$$-\frac{1}{2}(C_{(WX)} + \lambda_1 I)u^* + C_{(ZAX)(WX)}q_a - \mu_{(WX)} - \frac{1}{2}(C_{(WX)} + \lambda_1 I)u^* = 0$$

$$C_{(ZAX)(WX)}q_a - \mu_{(WX)} = (C_{(WX)} + \lambda_1 I)u^*.$$

This yields the optimal kernel embedded dual function $u^* \in H$

$$u^* = (C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}). \quad (13)$$

Putting back the u^* to the original function (12), we have

$$\begin{aligned} \min_{q_a \in F} \Gamma_{\lambda_1, \lambda_2}(q_a, u^*) &:= \langle C_{(ZAX)(WX)}q_a - \mu_{(WX)}, (C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}) \rangle_H \\ &\quad - \left\langle \frac{1}{2}C_{(WX)}(C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}), (C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}) \right\rangle_H \\ &\quad - \left\langle \frac{1}{2}\lambda_1(C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}), (C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}) \right\rangle_H \\ &\quad + \frac{1}{2}\lambda_2 \|q_a\|_F^2. \end{aligned}$$

Notice that

$$I - \frac{1}{2}C_{(WX)}(C_{(WX)} + \lambda_1 I)^{-1} - \frac{1}{2}\lambda_1(C_{(WX)} + \lambda_1 I)^{-1} = I - \frac{1}{2}(C_{(WX)} + \lambda_1 I)(C_{(WX)} + \lambda_1 I)^{-1} = \frac{1}{2}I.$$

Hence, the minimization problem is equivalent to

$$\min_{q_a \in F} \Gamma_{\lambda_1, \lambda_2}(q_a, u^*) := \left\langle \frac{1}{2}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}), (C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a - \mu_{(WX)}) \right\rangle_H + \frac{1}{2}\lambda_2 \|q_a\|_F^2. \quad (14)$$

By the uniqueness of minimizer q_a^* , we set the Fréchet derivative of the $\Gamma_{\lambda_1, \lambda_2}(q_a, u^*)$ at point $(q_a = q_a^*, u)$ to be zero and get

$$\begin{aligned} &\frac{1}{2}C_{(WX)(ZAX)}(C_{(WX)} + \lambda_1 I)^{-1}(C_{(ZAX)(WX)}q_a^* - \mu_{(WX)}) \\ &\quad + C_{(WX)(ZAX)}(C_{(WX)} + \lambda_1 I)^{-1}\frac{1}{2}(C_{(ZAX)(WX)}q_a^* - \mu_{(WX)}) + \lambda_2 q_a^* = 0 \\ \implies &(C_{(WX)(ZAX)}(C_{(WX)} + \lambda_1 I)^{-1}C_{(ZAX)(WX)} + \lambda_2 I)q_a^* = C_{(WX)(ZAX)}(C_{(WX)} + \lambda_1 I)^{-1}\mu_{(WX)}. \end{aligned}$$

This produces the optimal regularized kernel embedded solution $q_a^* \in F$

$$q_a^* = (C_{(WX)(ZAX)}(C_{(WX)} + \lambda_1 I)^{-1}C_{(ZAX)(WX)} + \lambda_2 I)^{-1}C_{(WX)(ZAX)}(C_{(WX)} + \lambda_1 I)^{-1}\mu_{(WX)}. \quad (15)$$

□

By virtue of the definition of universal kernels, the RKHS induced by the universal kernel is dense in the space of bounded continuous functions over a polish space. This means q_a^* (15) approximates to a bounded continuous function in $\mathcal{C}_0(Z, X)$ arbitrarily well.

2.2 Empirical form

The empirical version of the regularized kernel embedded solution $q_a^* \in F$ is given by

$$\hat{q}_a^* = (\hat{C}_{(WX)(ZAX)}(\hat{C}_{(WX)} + \lambda_1 I)^{-1} \hat{C}_{(ZAX)(WX)} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)}(\hat{C}_{(WX)} + \lambda_1 I)^{-1} \hat{\mu}_{(WX)}. \quad (16)$$

Since the Moore-Aronszajn theorem states that a RKHS and its reproducing kernel uniquely determine each other, and gives the form of elements in RKHSs induced by kernels on scattered samples, we can subsequently acquire the expressions of $q_a^* \in F$ and $u^* \in H$, where q_a^* is given by

$$q_a^* = \sum_{i \geq 1} \alpha_i \psi(z_i, a_i, x_i).$$

Usually we can approximate q_a^* by its orthogonal projection denoted by $P_n q_a^*$ on n -dim subspace of F spanned by feature maps on scattered points. This subspace is defined by $F_n := \{f : f = \sum_{i=1}^n \gamma_i \psi(z_i, a_i, x_i), \gamma_i \in \mathbb{R}\}$ and $P_n q_a^*$ has a expression of

$$P_n q_a^* = \sum_{i=1}^n \alpha_i \psi(z_i, a_i, x_i). \quad (17)$$

As $n \rightarrow \infty$, we have that $\|P_n q_a^* - q_a^*\|_F$ converges to 0 with exponential decay rate, which is faster than polynomial decay rate, if the kernel function is smooth enough, for example Gaussian RBF kernel. One can refer to [2, 15] for detailed proofs and derivations.

We deem all empirical (cross-)covariance embeddings here by operators between finite dimensional spaces. Then, the empirical version (16) approximates the $P_n q_a^*$ in F_n with kernel representor

$$\hat{q}_a^* = \sum_{i=1}^n \hat{\alpha}_i \psi(z_i, a_i, x_i). \quad (18)$$

The coefficients $(\hat{\alpha}_i)_{1 \leq i \leq n}$ should be represented by empirical embeddings. To explain the finite dimensional subspaces F_n and $H_n := \{h : h = \sum_{i=1}^n \zeta_i \phi(w_i, x_i), \zeta_i \in \mathbb{R}\}$, we delineate the relationship between empirical (cross-)covariance embeddings and Gram matrices.

Define operators $\Phi = (\phi(w_1, x_1), \dots, \phi(w_n, x_n)) : \mathbb{R}^n \mapsto H_n$ by $\Phi \beta = \sum_{i=1}^n \beta_i \phi(w_i, x_i), \forall \beta \in \mathbb{R}^n$ and its adjoint $\Phi^* : H_n \mapsto \mathbb{R}^n$ by $\Phi^* h = (\langle \phi(w_i, x_i), h \rangle_{H_n})_{i=1}^n = (h(w_i, x_i))_{i=1}^n$; $\Psi = (\psi(z_1, a_1, x_1), \dots, \psi(z_n, a_n, x_n)) = (\mathbb{1}_{a_1=a} \tilde{\psi}(z_1, x_1), \dots, \mathbb{1}_{a_n=a} \tilde{\psi}(z_n, x_n)) : \mathbb{R}^n \mapsto F_n$ and its adjoint $\Psi^* : F_n \mapsto \mathbb{R}^n$, where $(w_i, x_i, a_i, z_i)_{1 \leq i \leq n}$ are i.i.d samples from $P_{(W,X,A,Z|U)}$. Then without centering, the empirical mean embedding and empirical (cross-)covariance embeddings under the samples are given by

$$\begin{aligned} \hat{C}_{(WX)} &= \frac{1}{n} \sum_{i=1}^n \phi(w_i, x_i) \otimes \phi(w_i, x_i) = \frac{1}{n} \Phi \Phi^*, & \hat{\mu}_{(WX)} &= \frac{1}{n} \sum_{i=1}^n \phi(w_i, x_i) = \frac{1}{n} \Phi \mathbb{1}_n, \\ \hat{C}_{(WX)(ZAX)} &= \frac{1}{n} \sum_{i=1}^n (\mathbb{1}_{a_i=a} \tilde{\psi}(z_i, x_i)) \otimes \phi(w_i, x_i) = \frac{1}{n} \sum_{i=1}^n \psi(z_i, a_i, x_i) \otimes \phi(w_i, x_i) = \frac{1}{n} \Psi \Phi^*, \\ \hat{C}_{(ZAX)(WX)} &= \frac{1}{n} \sum_{i=1}^n \phi(w_i, x_i) \otimes (\mathbb{1}_{a_i=a} \tilde{\psi}(z_i, x_i)) = \frac{1}{n} \sum_{i=1}^n \phi(w_i, x_i) \otimes \psi(z_i, a_i, x_i) = \frac{1}{n} \Phi \Psi^*. \end{aligned}$$

To make it consistent with the definition of Φ and Ψ , we write (18) explicitly by

$$\hat{q}_a^* = \sum_{i=1}^n \hat{\alpha}_i \psi(z_i, a_i, x_i) = \Psi \hat{\Lambda}, \quad \hat{\Lambda} = (\hat{\alpha}_1, \dots, \hat{\alpha}_n)^T \in \mathbb{R}^n. \quad (19)$$

The following proposition illustrates the derivation of the expressions of $\hat{q}_a^* = \Psi \hat{\Lambda}$ and $\hat{u}^* = \Phi \hat{\Gamma}$.

Proposition 2.3 (Kernel representor of $\hat{q}_a^*$ and $\hat{u}^*$)

Denote the Gram matrices of kernels $k(\cdot, \cdot)$ and $l(\cdot, \cdot)$ by $K = \Phi^* \Phi$ and $L = \Psi^* \Psi$.¹ Then the empirical regularized kernel embedded solution $\hat{q}_a^* \in F_n$ and $\hat{u}^* \in H_n$ have respective kernel representor $\hat{q}_a^* = \Psi \hat{\Lambda}$ and $\hat{u}^* = \Phi \hat{\Gamma}$, where

¹We recognize them as matrices on $\mathbb{R}^n \times \mathbb{R}^n$ and operators from $\mathbb{R}^n$ to $\mathbb{R}^n$ at the same time.

the coefficient $\widehat{\Lambda}$ and $\widehat{\Gamma}$ are given by

$$\widehat{\Lambda} = (L(K + n\lambda_1 I_n)^{-1}KL + n\lambda_2 L)^{-1}L(K + n\lambda_1 I_n)^{-1}K\mathbb{1}_n, \quad (20)$$

$$\widehat{\Gamma} = (K + n\lambda_1 I)^{-1}(L\widehat{\Lambda} - \mathbb{1}_n). \quad (21)$$

Proof:

Substituting the kernel embeddings by their matrices versions, the empirical version (16) satisfies

$$\begin{aligned} \widehat{q}_a^* &= \left(\frac{1}{n}\Psi\Phi^*\left(\frac{1}{n}\Phi\Phi^* + \lambda_1 I\right)^{-1}\frac{1}{n}\Phi\Psi^* + \lambda_2 I\right)^{-1}\frac{1}{n}\Psi\Phi^*\left(\frac{1}{n}\Phi\Phi^* + \lambda_1 I\right)^{-1}\frac{1}{n}\Phi\mathbb{1}_n \\ &= (\Psi\Phi^*(\Phi\Phi^* + n\lambda_1 I)^{-1}\Phi\Psi^* + n\lambda_2 I)^{-1}\Psi\Phi^*(\Phi\Phi^* + n\lambda_1 I)^{-1}\Phi\mathbb{1}_n. \end{aligned}$$

Apply the operator $\Psi\Phi^*(\Phi\Phi^* + n\lambda_1 I)^{-1}\Phi\Psi^* + n\lambda_2 I$ to both sides on the left and get

$$\begin{aligned} (\Psi\Phi^*(\Phi\Phi^* + n\lambda_1 I)^{-1}\Phi\Psi^* + n\lambda_2 I)\widehat{q}_a^* &= \Psi\Phi^*(\Phi\Phi^* + n\lambda_1 I)^{-1}\Phi\mathbb{1}_n \\ (\Psi\Phi^*(\Phi\Phi^* + n\lambda_1 I)^{-1}\Phi\Psi^* + n\lambda_2 I)\Psi\widehat{\Lambda} &= \Psi\Phi^*(\Phi\Phi^* + n\lambda_1 I)^{-1}\Phi\mathbb{1}_n \quad (\widehat{q}_a^* = \Psi\widehat{\Lambda}) \\ (\Psi(\Phi^*\Phi + n\lambda_1 I)^{-1}\Phi^*\Phi\Psi^*\Psi + n\lambda_2 \Psi)\widehat{\Lambda} &= \Psi(\Phi^*\Phi + n\lambda_1 I)^{-1}\Phi^*\Phi\mathbb{1}_n. \end{aligned}$$

After left applying Ψ^* to the both sides, by the injection of Ψ^* , we get

$$\begin{aligned} (\Psi^*\Psi(\Phi^*\Phi + n\lambda_1 I)^{-1}\Phi^*\Phi\Psi^*\Psi + n\lambda_2 \Psi^*\Psi)\widehat{\Lambda} &= \Psi^*\Psi(\Phi^*\Phi + n\lambda_1 I)^{-1}\Phi^*\Phi\mathbb{1}_n \\ (L(K + n\lambda_1 I_n)^{-1}KL + n\lambda_2 L)\widehat{\Lambda} &= L(K + n\lambda_1 I_n)^{-1}K\mathbb{1}_n. \end{aligned}$$

If $L(K + n\lambda_1 I_n)^{-1}KL + n\lambda_2 L$ is invertible, this gives the kernel embedded coefficient

$$\widehat{\Lambda} = (L(K + n\lambda_1 I_n)^{-1}KL + n\lambda_2 L)^{-1}L(K + n\lambda_1 I_n)^{-1}K\mathbb{1}_n.$$

For $\widehat{\Gamma}$, we use the empirical version of (13).

$$\begin{aligned} \widehat{u}^* &= \Phi\widehat{\Gamma} = \left(\frac{1}{n}\Phi\Phi^* + \lambda_1 I\right)^{-1}\left(\frac{1}{n}\Phi\Psi^*\widehat{q}_a^* - \frac{1}{n}\Phi\mathbb{1}_n\right) \\ &= (\Phi\Phi^* + n\lambda_1 I)^{-1}(\Phi\Psi^*\Psi\widehat{\Lambda} - \Phi\mathbb{1}_n) \\ &= \Phi(\Phi^*\Phi + n\lambda_1 I)^{-1}(\Psi^*\Psi\widehat{\Lambda} - \mathbb{1}_n) \\ &= \Phi(K + n\lambda_1 I)^{-1}(L\widehat{\Lambda} - \mathbb{1}_n), \end{aligned}$$

which indicates that $\widehat{\Gamma} = (K + n\lambda_1 I)^{-1}(L\widehat{\Lambda} - \mathbb{1}_n)$ by the injection of Φ . □

Since $L(K + n\lambda_1 I_n)^{-1}KL + n\lambda_2 L$ is not guaranteed to be invertible or well-posed by the Gram matrix L , we can use the Tikhonov regularized solution instead, which is given by

$$\widehat{\Lambda}_\lambda := (C^*C + \lambda I)^{-1}C^*b, \quad (22)$$

where $C := L(K + n\lambda_1 I_n)^{-1}KL + n\lambda_2 L$, $b := L(K + n\lambda_1 I_n)^{-1}K\mathbb{1}_n$. (22) converges to the minimum-norm solution as $\lambda \rightarrow 0$.

Remark 2.1 *Of note, since $q_a \in \tilde{F} \cap F$, the coefficients of the linear expansion of q_a with respect to the feature maps in F and $\tilde{F}$ have the following matching rule: i) the coefficient α_i is shared in both spaces if $a_i = a$ and ii) the coefficient α_i given by F is 0 if $a_i \neq a$.² However, if going back to the Tikhonov regularized solution (22) we have derived, it might not strictly follows the matching rule, since its norm must be minimal among all the solutions and hence is probably not the true kernel representer (20) that we want. This means the computed Tikhonov regularized solution (22) in numerical experiments has risks to be scaled by a scalar compared to the true solution. To decrease the risk of the unwanted results, one can divide the sample set by treatment and compute $E[Y^a]$ using the corresponding samples with an estimator, but not using the whole sample set to compute $E[Y^1]$ and $E[Y^0]$ at the same time.*

²Consider that only samples with treatment a is included in $\tilde{F}$.

2.3 Tikhonov regularized kernel embedded outcome confounding bridge function, empirical form

In this part we use dual kernel embedding method to give a closed-form expression of outcome confounding bridge function on an RKHS and its empirical form. We follow the logic as deriving the kernel embedded estimator of treatment confounding bridge function and write the derivation with the simplest description.

Part i) integral equation transformation:

$$\begin{aligned} E[Y|Z, A=a, X] &= E[h_a(W, a, X)|Z, A=a, X] \\ f(a|Z, X)E[Y|Z, A=a, X] &= f(a|Z, X)E[h_a(W, a, X)|Z, A=a, X] \\ E[\mathbb{1}_{A=a}Y|Z, X] &= E[\mathbb{1}_{A=a}h_a(W, a, X)|Z, X]. \end{aligned}$$

Part ii) ERM reformulation, minimax problem reformulation:

$$\begin{aligned} &\min_{h_a \in L_2(P_{(W, X|Z, X)})} E[l(E[\mathbb{1}_{A=a}Y|Z, X], E[\mathbb{1}_{A=a}h_a(W, a, X)|Z, X])] \\ &= \min_{h_a \in L_2(P_{(W, X|Z, X)})} E[\max_{u \in \mathbb{R}} \{uE[\mathbb{1}_{A=a}h_a(W, a, X)|Z, X] - uE[\mathbb{1}_{A=a}Y|Z, X] - \frac{u^2}{2}\}] \\ &= \min_{h_a \in L_2(P_{(W, X|Z, X)})} \max_{u \in \mathcal{M}(Z, X)} E[u(Z, X)E[\mathbb{1}_{A=a}h_a(W, a, X)|Z, X] - u(Z, X)E[\mathbb{1}_{A=a}Y|Z, X] - \frac{1}{2}u^2(Z, X)] \\ &= \min_{h_a \in L_2(P_{(W, X|Z, X)})} \max_{u \in \mathcal{M}(Z, X)} E[u(Z, X)\mathbb{1}_{A=a}h_a(W, a, X) - u(Z, X)\mathbb{1}_{A=a}Y - \frac{1}{2}u^2(Z, X)] \\ &= \min_{h_a \in \mathcal{C}_0(W, X)} \max_{u \in \mathcal{C}_0(Z, X)} E[u(Z, X)\mathbb{1}_{A=a}h_a(W, a, X) - u(Z, X)\mathbb{1}_{A=a}Y - \frac{1}{2}u^2(Z, X)]. \end{aligned}$$

Part iii) Kernel embedding of means and cross-covariances:

$$\begin{aligned} &\min_{h_a \in G} \max_{u \in D} E[u(Z, X)\mathbb{1}_{A=a}h_a(W, a, X) - u(Z, X)\mathbb{1}_{A=a}Y - \frac{1}{2}u^2(Z, X)] \\ &= \min_{h_a \in G} \max_{u \in D} E[\langle u, \rho(Z, X) \rangle_D \mathbb{1}_{A=a} \langle h_a, \tilde{\pi}(W, X) \rangle_{\tilde{G}} - \langle u, \rho(Z, X) \rangle_D \mathbb{1}_{A=a}Y - \frac{1}{2} \langle C_{(ZX)}u, u \rangle_D] \\ &= \min_{h_a \in G} \max_{u \in D} E[\langle u, \rho(Z, X) \rangle_D \langle h_a, \pi(W, A, X) \rangle_G - \langle u, \mathbb{1}_{A=a}Y\rho(Z, X) \rangle_D - \frac{1}{2} \langle C_{(ZX)}u, u \rangle_D] \\ &= \min_{h_a \in G} \max_{u \in D} E[\langle u \otimes h_a, \rho(Z, X) \otimes \pi(W, A, X) \rangle_{HS(G, D)} - \langle u, \mathbb{1}_{A=a}Y\rho(Z, X) \rangle_D - \frac{1}{2} \langle C_{(ZX)}u, u \rangle_D] \\ &= \min_{h_a \in G} \max_{u \in D} \langle u \otimes h_a, C_{(WAX)(ZX)} \rangle_{HS(G, D)} - \langle u, E[\mathbb{1}_{A=a}Y\rho(Z, X)] \rangle_D + \frac{1}{2} \langle C_{(ZX)}u, u \rangle_D \\ &= \min_{h_a \in G} \max_{u \in D} \langle u, C_{(WAX)(ZX)}h_a - E[\mathbb{1}_{A=a}Y\rho(Z, X)] - \frac{1}{2}C_{(ZX)}u \rangle_D. \end{aligned}$$

Part iv) Tikhonov regularized solutions:

$$\begin{aligned} &\min_{h_a \in G} \max_{u \in D} \langle u, C_{(WAX)(ZX)}h_a - E[\mathbb{1}_{A=a}Y\rho(Z, X)] - \frac{1}{2}C_{(ZX)}u \rangle_D + \frac{\lambda_2}{2} \langle h_a, h_a \rangle_G - \frac{\lambda_1}{2} \langle u, u \rangle_D. \\ &\begin{cases} u^* = (C_{(ZX)} + \lambda_1 I)^{-1} (C_{(WAX)(ZX)}h_a - E[\mathbb{1}_{A=a}Y\rho(Z, X)]), \\ h_a^* = (C_{(ZX)(WAX)}(C_{(ZX)} + \lambda_1 I)^{-1} C_{(WAX)(ZX)} + \lambda_2 I)^{-1} C_{(ZX)(WAX)}(C_{(ZX)} + \lambda_1 I)^{-1} E[\mathbb{1}_{A=a}Y\rho(Z, X)]. \end{cases} \end{aligned}$$

Part v) Empirical forms:

$$\begin{aligned} \hat{h}_a^* &= \Pi \hat{\Lambda} = \Pi(L(K + n\lambda_1 I)^{-1}KL + n\lambda_2 L)^{-1}L(K + n\lambda_1 I)^{-1}K(\mathbb{1}_{A=a}Y), \\ \hat{u}^* &= \rho \hat{\Gamma} = \rho(K + n\lambda_1 I)^{-1}(L\hat{\Lambda} - \mathbb{1}_{A=a}Y), \end{aligned}$$

where $\rho = (\rho(z_1, x_1), \dots, \rho(z_n, x_n))$, $\Pi = (\mathbb{1}_{a_1=a}\tilde{\pi}(w_1, x_1), \dots, \mathbb{1}_{a_n=a}\tilde{\pi}(w_n, x_n)) = (\pi(w_1, a_1, x_1), \dots, \pi(w_n, a_n, x_n))$, $K = \rho^* \rho$, $L = \Pi^* \Pi$.

2.4 Analysis on convergence of the bridge function estimator

Since RKHS F is dense in $\mathcal{C}_0(Z, X)$, under Assumption 2.1, there exist a RKHS function that approximate the true bounded continuous treatment confounding bridge function arbitrarily well. We denote by this function $q_a^\dagger$. $q_a^\dagger \in F$ is a necessary source condition for the convergence analysis. Now we have the decomposition of the difference between $\hat{q}_a^*$ and $q_a^\dagger$ in RKHS norm by

$$\|\hat{q}_a^* - q_a^\dagger\|_F \leq \|q_a^* - q_a^\dagger\|_F + \|\hat{q}_a^* - P_n q_a^*\|_{F_n} + \|P_n q_a^* - q_a^*\|_F. \quad (23)$$

As mentioned below (17), and the polynomial decay rate due to $\sqrt{n}$ -convergence of empirical kernel embeddings, the rate of convergence of the last term on the right hand side of (23) is far faster than the first two terms if the kernel is smooth enough. Therefore, in the following parts, we derive the decay rate of $\|q_a^* - q_a^\dagger\|_F$ and $\|\hat{q}_a^* - P_n q_a^*\|_{F_n}$ respectively.

2.4.1 Convergence of the regularized form to the RKHS approximator

Lemma 2.4 (*Kernel embedded form of the original problem*)

If Assumption 2.1 holds, then $q_a^\dagger \in F$ approximates the true bounded continuous solution of (4) as a solution of

$$C_{(ZAX)(WX)} q_a = \mu_{(WX)}.$$

Proof:

For any bounded continuous solution solving the original integral equation (4), for all $g \in H$, we have

$$\begin{aligned} E_{W,X,A,Z}[\mathbb{1}_{A=a} q_a(Z, a, X) g(W, X)] &= E_{W,X} E_{A,Z}[\mathbb{1}_{A=a} q_a(Z, a, X) g(W, X) | W, X] \\ &= E_{W,X} [g(W, X) \underbrace{E_{A,Z}[\mathbb{1}_{A=a} q_a(Z, a, X) | W, X]}_{=1 \text{ by (4)}}] \\ &= E_{W,X} [g(W, X)]. \end{aligned} \quad (24)$$

Hence, the bounded continuous solution must be a solution of (24) as well. Moreover, if Assumption 2.1 holds, $q_a^\dagger$ approximates the bounded continuous solution arbitrarily well and satisfies

$$\left\langle C_{(ZAX)(WX)} q_a^\dagger, g \right\rangle_H = \langle \mu_{(WX)}, g \rangle_H,$$

which implies that

$$C_{(ZAX)(WX)} q_a^\dagger = \mu_{(WX)}. \quad (25)$$

□

Given a positive constant that is large enough to make $C_{(WX)} + \lambda_1 I$ boundedly invertible, we know

$$(C_{(WX)} + \lambda_1 I)^{-\frac{1}{2}} C_{(ZAX)(WX)} q_a = (C_{(WX)} + \lambda_1 I)^{-\frac{1}{2}} \mu_{(WX)} \quad (26)$$

and (25) shares the same set of solutions. We simplify $(C_{(WX)} + \lambda_1 I)^{-\frac{1}{2}} C_{(ZAX)(WX)}$, $(C_{(WX)} + \lambda_1 I)^{-\frac{1}{2}} \mu_{(WX)}$ by A_{λ_1} , μ_{λ_1} for the next theorem.

Theorem 2.2 (*Convergence of the regularized kernel embedded solution*)

Given the positive regularization parameter λ_2 , then as $\lambda_2 \rightarrow 0$, the regularized kernel embedded solution q_a^* (15) converges to $q_a^\dagger$ as a solution of preconditioned ill-posed problem (26) contained in F .

Proof:

First we notice that q_a^* can be written by $A_{\lambda_1}^* A_{\lambda_1} + \lambda_2 I$. By spectral theorem of compact operators[9], the operators $A_{\lambda_1}^* A_{\lambda_1} + \lambda_2 I$ and A_{λ_1} have spectral decompositions

$$A_{\lambda_1}^* A_{\lambda_1} + \lambda_2 I = \sum_{i \geq 1} (\sigma_i^2 + \lambda_2) f_i \otimes f_i \quad (27)$$

$$A_{\lambda_1} = \sum_{i \geq 1} \sigma_i h_i \otimes f_i, \quad (28)$$

where the singular values $(\sigma_i)_{i \geq 1}$ of A_{λ_1} with corresponding left and right eigenvectors $(f_i)_{i \geq 1}$, $(h_i)_{i \geq 1}$ form the singular system of A_{λ_1} , denoted by $(\sigma_i, h_i, f_i)_{i \geq 1}$. Moreover, $(f_i)_{i \geq 1}$ and $(h_i)_{i \geq 1}$ form orthonormal bases in F and H respectively.

Based on the previous statements, we have that the solution $q_a^\dagger$ and kernel embedded solution q_a^* satisfy

$$A_{\lambda_1} q_a^\dagger = \mu_{\lambda_1} \quad (29)$$

$$(A_{\lambda_1}^* A_{\lambda_1} + \lambda_2 I) q_a^* = A_{\lambda_1}^* \mu_{\lambda_1}, \quad (30)$$

Replacing the operator A_{λ_1} in (29) by its spectral decomposition (28), we have

$$\begin{aligned} \mu_{\lambda_1} &= \sum_{i \geq 1} \sigma_i (h_i \otimes f_i) (q_a^\dagger) \\ &= \sum_{i: \sigma_i \neq 0}^{\infty} \sigma_i \langle q_a^\dagger, f_i \rangle_F h_i \\ \implies \langle h_i, \mu_{\lambda_1} \rangle_H &= \sigma_i \langle q_a^\dagger, f_i \rangle_F \\ \frac{\langle h_i, \mu_{\lambda_1} \rangle_H}{\sigma_i} &= \langle q_a^\dagger, f_i \rangle_F. \end{aligned}$$

Since $q_a^\dagger \in F$, this implies that

$$q_a^\dagger = \sum_{i: \sigma_i \neq 0}^{\infty} \frac{\langle h_i, \mu_{\lambda_1} \rangle_H}{\sigma_i} f_i. \quad (31)$$

Replacing the operator $A_{\lambda_1}^* A_{\lambda_1} + \lambda_2 I$ in (30) by its spectral decomposition (27), we have

$$\begin{aligned} A_{\lambda_1}^* \mu_{\lambda_1} &= \sum_{i \geq 1} (\sigma_i^2 + \lambda_2) (f_i \otimes f_i) (q_a^*) \\ &= \sum_{i \geq 1} (\sigma_i^2 + \lambda_2) \langle f_i, q_a^* \rangle_F f_i \\ \implies \langle f_i, A_{\lambda_1}^* \mu_{\lambda_1} \rangle_F &= (\sigma_i^2 + \lambda_2) \langle f_i, q_a^* \rangle_F \\ \langle A_{\lambda_1} f_i, \mu_{\lambda_1} \rangle_H &= (\sigma_i^2 + \lambda_2) \langle f_i, q_a^* \rangle_F \\ \left\langle \sum_{j \geq 1} \sigma_j (h_j \otimes f_j) (f_i), \mu_{\lambda_1} \right\rangle_H &= (\sigma_i^2 + \lambda_2) \langle f_i, q_a^* \rangle_F \\ \langle \sigma_i h_i, \mu_{\lambda_1} \rangle_H &= (\sigma_i^2 + \lambda_2) \langle f_i, q_a^* \rangle_F \\ \frac{\sigma_i \langle h_i, \mu_{\lambda_1} \rangle_H}{\sigma_i^2 + \lambda_2} &= \langle f_i, q_a^* \rangle_F. \end{aligned}$$

By the closed-form of q_a^* (15), it is an element in F . So, its decomposition in F is given by

$$q_a^* = \sum_{i \geq 1} \frac{\sigma_i \langle h_i, \mu_{\lambda_1} \rangle_H}{\sigma_i^2 + \lambda_2} f_i = \sum_{i: \sigma_i \neq 0}^{\infty} \frac{\sigma_i \langle h_i, \mu_{\lambda_1} \rangle_H}{\sigma_i^2 + \lambda_2} f_i. \quad (32)$$

Hence, as $\lambda_2 \rightarrow 0$, the difference of the two solutions in RKHS norm is given by

$$\begin{aligned}
\|q_a^* - q_a^\dagger\|_F &= \left\| \sum_{i:\sigma_i \neq 0} \frac{\sigma_i \langle h_i, \mu_{\lambda_1} \rangle_H}{\sigma_i^2 + \lambda_2} f_i - \sum_{i:\sigma_i \neq 0} \frac{\langle h_i, \mu_{\lambda_1} \rangle_H}{\sigma_i} f_i \right\|_F \\
&= \left\| \sum_{i:\sigma_i \neq 0} \langle h_i, \mu_{\lambda_1} \rangle_H f_i \left(\frac{\sigma_i}{\sigma_i^2 + \lambda_2} - \frac{1}{\sigma_i} \right) \right\|_F \\
&= \left\| \sum_{i:\sigma_i \neq 0} \langle h_i, \mu_{\lambda_1} \rangle_H f_i \left(\frac{\lambda_2}{\sigma_i^3 + \lambda_2 \sigma_i} \right) \right\|_F \\
&= \left\| \sum_{i:\sigma_i \neq 0} \langle h_i, \mu_{\lambda_1} \rangle_H f_i \left(\frac{1}{\lambda_2^{-1} \sigma_i^3 + \sigma_i} \right) \right\|_F \\
&= \sqrt{\sum_{i:\sigma_i \neq 0} \frac{\langle h_i, \mu_{\lambda_1} \rangle_H^2}{(\lambda_2^{-1} \sigma_i^3 + \sigma_i)^2}}.
\end{aligned}$$

Denote the spectral expansion of μ_{λ_1} in H by $\mu_{\lambda_1} = \sum_{i \geq 1} c_i h_i$, where $\sum_{i \geq 1} c_i^2 < \infty$ by its finite square RKHS norm. Then the difference of the two solutions in RKHS norm is upper bounded by

$$\|q_a^* - q_a^\dagger\|_F^2 = \sum_{i:\sigma_i \neq 0} \frac{\langle h_i, \mu_{\lambda_1} \rangle_H^2}{(\lambda_2^{-1} \sigma_i^3 + \sigma_i)^2} = \sum_{i:\sigma_i \neq 0} \frac{c_i^2}{\sigma_i^2} \frac{1}{(\lambda_2^{-1} \sigma_i^2 + 1)^2}. \quad (33)$$

Since $q_a^\dagger \in F$ implies the finite square RKHS norm of $q_a^\dagger$, its decomposition in F gives $\sum_{i:\sigma_i \neq 0} \frac{c_i^2}{\sigma_i^2} < \infty$. Combining $\lim_{\lambda_2 \rightarrow 0} \frac{1}{(\lambda_2^{-1} \sigma_i^2 + 1)^2} = 0$ and $0 < \frac{c_i^2}{\sigma_i^2} \frac{1}{(\lambda_2^{-1} \sigma_i^2 + 1)^2} < \frac{c_i^2}{\sigma_i^2}$ for any $i : \sigma_i \neq 0$, we have (33) asymptotically converges to 0 as $\lambda_2 \rightarrow 0$. $\square$

Remark 2.2 To get the convergence rate of q_a^* to $q_a^\dagger$ in RKHS norm, we need to focus on the decay rate of the difference between orthogonal projections onto F_n , i.e. $\|P_n q_a^* - P_n q_a^\dagger\|_{F_n}$. In fact, this rate is determined by the stabilizer λ_1 (irrelevant to n) and the behavior of covariance embedding $C_{(WX)(ZAX)}$. If the λ_1 is large enough to make $C_{(WX)} + \lambda_1 I$ boundedly invertible, and $C_{(WX)(ZAX)}$ is well enough to bound the singular values away from zero, which happens only on finite dimensional spaces, then there exist a positive constant l such that $\frac{1}{\sigma_i} \leq \frac{1}{l}$, $1 \leq i \leq n$. If the kernel function is smooth enough such that $\|q_a^* - P_n q_a^*\|_F$ and $\|q_a^\dagger - P_n q_a^\dagger\|_F$ decay exponentially about n , (33) is upper bounded by $\frac{1}{l^2} \sum_{i=1}^n c_i^2 \frac{1}{(\lambda_2^{-1} l^2 + 1)^2} + \mathcal{O}(e^{-n}) \asymp \lambda_2^2$ as $\lambda_2 \rightarrow 0$.

Remark 2.3 A worse case is mild ill-posedness that singular values of A_{λ_1} polynomially decay to 0 by $\sigma_i \asymp i^{-u}$. Lemma 8.1 in [6] discusses the asymptotic behavior of (33) depending on the source condition that $\xi = (\langle h_i, \mu_{\lambda_1} \rangle_H)_{i \geq 1}$ belongs to a weighted l_2 space S^q for some $q \geq 0$. However the rate of convergence in this situation is still difficult to find because the inequality technique used by Lemma 8.1 usually leads to a divergent upper bound dominant by λ_2^{-1} which goes to infinity. If we choose $\lambda_2^{-1} \rightarrow 0$ as $n \rightarrow \infty$, like setting $\lambda_2 = n^{-\beta}$ for $\beta < 0$, then $\|q_a^* - q_a^\dagger\|_F$ asymptotically converges to 0. Of note $\hat{q}_a^*$ and q_a^* are also drawn to 0 functionals in F by the regularization of extremely large λ_2 , which result in a false convergence. One can turn to (35) in Theorem 2.3 to check the influence of λ_2 on the spectrum.

2.4.2 Convergence of the empirical form to the regularized form

Theorem 2.3 (Asymptotic convergence of the empirical kernel embedded solution)

Suppose the universal kernel functions $k(\cdot, \cdot)$ and $l(\cdot, \cdot)$ are bounded and smooth enough. Given regularization parameter λ_1 which is a fixed positive number and $\lambda_2 = n^{-\beta}$ with $\beta > 0$, the empirical regularized kernel embedded solution $\hat{q}_a^*$ converges to the regularized kernel embedded solution q_a^* of the regularized task (12) in RKHS norm with convergence rate

$$E \|\hat{q}_a^* - q_a^*\|_{F_n} \leq \begin{cases} \mathcal{O}(n^{2\beta - \frac{1}{2}}), & \text{if } \inf_{\lambda} \{\lambda \in \sigma(\hat{R}_{\lambda_1})\} = \inf_{\lambda} \{\lambda \in \sigma(R_{\lambda_1})\} = 0, \\ \mathcal{O}(n^{-\frac{1}{2}}), & \text{if } \inf_{\lambda} \{\lambda \in \sigma(\hat{R}_{\lambda_1})\} > 0 \text{ and } \inf_{\lambda} \{\lambda \in \sigma(R_{\lambda_1})\} > 0. \end{cases}$$

Proof:

By our assumption that the kernel functions are smooth enough, the convergence $E\|P_n q_a^* - q_a^*\|_{F_n}$ is exponentially fast and is faster than the polynomial decay rate of $E\|\hat{q}_a^* - P_n q_a^*\|$. Since $E\|\hat{q}_a^* - q_a^*\|_{F_n} \leq E\|\hat{q}_a^* - P_n q_a^*\|_{F_n} + E\|P_n q_a^* - q_a^*\|_{F_n}$, we focus on the upper bound of the decay rate of $E\|\hat{q}_a^* - P_n q_a^*\|_{F_n}$.

For simplicity, we continue to use the notations given by Muandet et al. [8],

$$\begin{aligned} C_{\lambda_1} &= C_{(WX)} + \lambda_1 I, & R_{\lambda_1} &= C_{(WX)(ZAX)} C_{\lambda_1}^{-1} C_{(ZAX)(WX)}, \\ \hat{C}_{\lambda_1} &= \hat{C}_{(WX)} + \lambda_1 I, & \hat{R}_{\lambda_1} &= \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{C}_{(ZAX)(WX)}. \end{aligned}$$

We don't introduce new notations of orthogonal projections of (cross-)covariance and mean embeddings onto F_n for simplicity. Then the expectation of difference of $\hat{q}_a^*$ and $P_n q_a^*$ in RKHS norm is given by

$$\begin{aligned} E\|\hat{q}_a^* - P_n q_a^*\|_{F_n} &\leq E\|\hat{q}_a^* - P_n q_a^*\|_{F_n} + E\|P_n q_a^* - q_a^*\|_{F_n} \\ &= E\|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)} - (R_{\lambda_1} + \lambda_2 I)^{-1} C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \mu_{(WX)}\|_{F_n} \\ &\leq E\|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)} - (R_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)}\|_{F_n} \\ &\quad + E\|(R_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)} - (R_{\lambda_1} + \lambda_2 I)^{-1} C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \mu_{(WX)}\|_{F_n} \\ &=: E[T_1] + E[T_2]. \end{aligned}$$

Part i) Bounding $E[T_1]$:

$$\begin{aligned} E[T_1] &= E\left[\|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)} - (R_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)}\|_{F_n}\right] \\ &\leq E\left[\|\hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)}\|_{F_n} \|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} - (R_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F)}\right] \\ &= E\left[\|\hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)}\|_{F_n} \|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} (\hat{R}_{\lambda_1} + \lambda_2 I) ((\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} - (R_{\lambda_1} + \lambda_2 I)^{-1})\|_{\mathcal{L}(F)}\right] \\ &= E\left[\|\hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)}\|_{F_n} \|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} (R_{\lambda_1} - \hat{R}_{\lambda_1}) (R_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F)}\right] \\ &\leq E\left[\|\hat{C}_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)} \|\hat{\mu}_{(WX)}\|_{F_n} \|\hat{C}_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F)} \right. \\ &\quad \left. \cdot \|(R_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F)} \|R_{\lambda_1} - \hat{R}_{\lambda_1}\|_{\mathcal{L}(F)}\right]. \end{aligned} \tag{34}$$

By the assumption that kernels are bounded, we know the reproducing property implies the feature maps are bounded in RKHS norms. We denote the upper bound for $k(\cdot, \cdot)$ over $(\mathcal{W} \times \mathcal{X}) \times (\mathcal{W} \times \mathcal{X})$ by t_ϕ^2 and the upper bound for $l(\cdot, \cdot)$ over $(\mathcal{Z} \times \mathcal{A} \times \mathcal{X}) \times (\mathcal{Z} \times \mathcal{A} \times \mathcal{X})$ by t_ψ^2 . Then, the upper bounds for the feature maps in RKHS norms are given by

$$\begin{aligned} \|\phi(W, X)\|_H &= \sqrt{k((W, X), (W, X))} \leq t_\phi, \quad \forall (W, X) \in (\mathcal{W} \times \mathcal{X}), \\ \|\psi(Z, A, X)\|_{F_n} &= \sqrt{l((Z, A, X), (Z, A, X))} \leq t_\psi, \quad \forall (Z, A, X) \in (\mathcal{Z} \times \mathcal{A} \times \mathcal{X}). \end{aligned}$$

Hence

$$\begin{aligned} \|\hat{C}_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)} &= \|\hat{C}_{(WX)(ZAX)}\|_{HS(H_n, F_n)} \leq \frac{1}{n} \sum_{i=1}^n \|\phi(w_i, x_i)\|_{H_n} \|\psi(z_i, a_i, x_i)\|_{F_n} \leq t_\phi t_\psi, \\ \|C_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)} &= \|E[\phi(W, X) \otimes \psi(Z, A, X)]\|_{HS(H_n, F_n)} \leq E\|\phi(W, X) \otimes \psi(Z, A, X)\|_{HS(H_n, F_n)} \\ &= E[\|\phi(W, X)\|_{H_n} \|\psi(Z, A, X)\|_{F_n}] \leq t_\phi t_\psi, \\ \|\hat{\mu}_{(WX)}\|_{H_n} &\leq \frac{1}{n} \sum_{i=1}^n \|\phi(w_i, x_i)\|_{H_n} \leq t_\phi. \end{aligned}$$

Next is to bound the operator norm of the inverse regularized operators. Recall the norm of self-adjoint operator is chosen as the maximum between the absolute values of the infimum of spectrum and maximum of spectrum[9].

Since all the inverse operators considered here are positive (semi-)definite, their spectra are non-negative. So, their operator norms are just the the supremum of the spectrum.

$$\begin{aligned}
\|\hat{C}_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} &\leq \frac{1}{\inf_{\lambda}\{\lambda \in \sigma(\hat{C}_{(WX)} + \lambda_1 I)\}} = \frac{1}{\inf_{\lambda}\{\lambda \in \sigma(\hat{C}_{WX})\} + \lambda_1} \leq \frac{1}{\lambda_1}, \\
\|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F)} &\leq \frac{1}{\inf_{\lambda}\{\lambda \in \sigma(\hat{R}_{\lambda_1} + \lambda_2 I)\}} = \frac{1}{\inf_{\lambda}\{\lambda \in \sigma(\hat{R}_{\lambda_1})\} + \lambda_2} = \frac{1}{\delta' + \lambda_2}, \\
\|(R_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F)} &\leq \frac{1}{\inf_{\lambda}\{\lambda \in \sigma(R_{\lambda_1} + \lambda_2 I)\}} = \frac{1}{\inf_{\lambda}\{\lambda \in \sigma(R_{\lambda_1})\} + \lambda_2} \leq \frac{1}{\delta + \lambda_2}.
\end{aligned} \tag{35}$$

Last, we have

$$\begin{aligned}
\|R_{\lambda_1} - \hat{R}_{\lambda_1}\|_{\mathcal{L}(F)} &= \|C_{(WX)(ZAX)}C_{\lambda_1}^{-1}C_{(ZAX)(WX)} - \hat{C}_{(WX)(ZAX)}\hat{C}_{\lambda_1}^{-1}\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F)} \\
&\leq \|C_{(WX)(ZAX)}C_{\lambda_1}^{-1}C_{(ZAX)(WX)} - \hat{C}_{(WX)(ZAX)}C_{\lambda_1}^{-1}\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F)} \\
&\quad + \|\hat{C}_{(WX)(ZAX)}C_{\lambda_1}^{-1}\hat{C}_{(ZAX)(WX)} - \hat{C}_{(WX)(ZAX)}\hat{C}_{\lambda_1}^{-1}\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F)} \\
&\leq \|C_{(WX)(ZAX)}C_{\lambda_1}^{-1}C_{(ZAX)(WX)} - C_{(WX)(ZAX)}C_{\lambda_1}^{-1}\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F)} \\
&\quad + \|C_{(WX)(ZAX)}C_{\lambda_1}^{-1}\hat{C}_{(ZAX)(WX)} - \hat{C}_{(WX)(ZAX)}C_{\lambda_1}^{-1}\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F)} \\
&\quad + \|\hat{C}_{(WX)(ZAX)}(C_{\lambda_1}^{-1} - \hat{C}_{\lambda_1}^{-1})\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F)}.
\end{aligned} \tag{36}$$

Notice that

$$C_{\lambda_1}^{-1} - \hat{C}_{\lambda_1}^{-1} = C_{\lambda_1}^{-1}(\hat{C}_{\lambda_1} - C_{\lambda_1})\hat{C}_{\lambda_1}^{-1} = C_{\lambda_1}^{-1}(\hat{C}_{WX} - C_{WX})\hat{C}_{\lambda_1}^{-1}. \tag{37}$$

Applying (37) to (36), we get

$$\begin{aligned}
E\|R_{\lambda_1} - \hat{R}_{\lambda_1}\|_{\mathcal{L}(F)} &\leq E \left[\|C_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)} \|C_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|\hat{C}_{(ZAX)(WX)} - C_{(ZAX)(WX)}\|_{HS(F_n, H_n)} \right. \\
&\quad + \|C_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F_n, H_n)} \|\hat{C}_{(WX)(ZAX)} - C_{(WX)(ZAX)}\|_{HS(H_n, F_n)} \\
&\quad + \|\hat{C}_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)} \|C_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|\hat{C}_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|\hat{C}_{(ZAX)(WX)}\|_{\mathcal{L}(F_n, H_n)} \|\hat{C}_{WX} - C_{WX}\|_{HS(H_n)} \Big] \\
&\leq E \left[\frac{t_{\phi} t_{\psi}}{\lambda_1} \|\hat{C}_{(ZAX)(WX)} - C_{(ZAX)(WX)}\|_{HS(F_n, H_n)} + \frac{t_{\phi} t_{\psi}}{\lambda_1} \|\hat{C}_{(WX)(ZAX)} - C_{(WX)(ZAX)}\|_{HS(H_n, F_n)} \right. \\
&\quad \left. + \frac{t_{\phi}^2 t_{\psi}^2}{\lambda_1^2} \|\hat{C}_{WX} - C_{WX}\|_{HS(H_n)} \right].
\end{aligned} \tag{38}$$

By the $\sqrt{n}$ -consistency of cross-covariance embedding, (38) is $\mathcal{O}(\frac{1}{\sqrt{n}})$. Hence, combined with the upper bounds of other terms in (34), the convergence rate of $E[T_1]$ is $\mathcal{O}(\frac{1}{(\delta' + \lambda_2)(\delta + \lambda_2)\sqrt{n}})$.

Part ii) Bounding T_2 :

$$\begin{aligned}
E[T_2] &= E\|(R_{\lambda_1} + \lambda_2 I)^{-1} \widehat{C}_{(WX)(ZAX)} \widehat{C}_{\lambda_1}^{-1} \widehat{\mu}_{(WX)} - (R_{\lambda_1} + \lambda_2 I)^{-1} C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \mu_{(WX)}\|_{F_n} \\
&\leq E\left[\frac{1}{\delta + \lambda_2} \|\widehat{C}_{(WX)(ZAX)} \widehat{C}_{\lambda_1}^{-1} \widehat{\mu}_{(WX)} - C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \mu_{(WX)}\|_{F_n}\right] \quad (\text{By 35}) \\
&\leq E\left[\frac{1}{\delta + \lambda_2} \|\widehat{C}_{(WX)(ZAX)} \widehat{C}_{\lambda_1}^{-1} \widehat{\mu}_{(WX)} - C_{(WX)(ZAX)} \widehat{C}_{\lambda_1}^{-1} \widehat{\mu}_{(WX)}\|_{F_n} \right. \\
&\quad \left. + \frac{1}{\delta + \lambda_2} \|C_{(WX)(ZAX)} \widehat{C}_{\lambda_1}^{-1} \widehat{\mu}_{(WX)} - C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \mu_{(WX)}\|_{F_n} \right. \\
&\quad \left. + \frac{1}{\delta + \lambda_2} \|C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \widehat{\mu}_{(WX)} - C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \mu_{(WX)}\|_{F_n}\right] \\
&\leq E\left[\frac{1}{\delta + \lambda_2} \|\widehat{C}_{(WX)(ZAX)} - C_{(WX)(ZAX)}\|_{HS(H_n, F_n)} \|\widehat{C}_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|\widehat{\mu}_{(WX)}\|_{F_n} \right. \\
&\quad \left. + \frac{1}{\delta + \lambda_2} \|C_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)} \|C_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|(\widehat{C}_{(WX)} - C_{(WX)})\|_{HS(H_n, F_n)} \|\widehat{C}_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|\widehat{\mu}_{(WX)}\|_{F_n} \right. \\
&\quad \left. + \frac{1}{\delta + \lambda_2} \|C_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)} \|C_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)} \|\widehat{\mu}_{(WX)} - \mu_{(WX)}\|_{F_n}\right] \quad (\text{By 37}) \\
&\leq E\left[\frac{t_\phi}{\lambda_1(\delta + \lambda_2)} \|\widehat{C}_{(WX)(ZAX)} - C_{(WX)(ZAX)}\|_{HS(H_n, F_n)} + \frac{t_\phi^2 t_\psi}{\lambda_1^2(\delta + \lambda_2)} \|(\widehat{C}_{(WX)} - C_{(WX)})\|_{HS(H_n, F_n)} \right. \\
&\quad \left. + \frac{t_\phi t_\psi}{\lambda_1(\delta + \lambda_2)} \|\widehat{\mu}_{(WX)} - \mu_{(WX)}\|_{F_n}\right]. \quad (39)
\end{aligned}$$

By the $\sqrt{n}$ -consistency of mean embedding and cross-covariance embedding, the convergence rate of $E[T_2]$ is $\mathcal{O}(\frac{1}{(\delta + \lambda_2)\sqrt{n}})$.

Part iii) Asymptotic convergence behavior of $\|\widehat{q}_a^* - P_n q_a^*\|_{F_n}$:

The convergence rates of $E[T_1]$ and $E[T_2]$ produce the convergence rate of $\|\widehat{q}_a^* - q_a^*\|_{F_n}$, which is given by

$$\mathcal{O}_p\left(\frac{1}{\sqrt{n}}\left(\frac{1}{\delta + \lambda_2} + \frac{1}{(\delta' + \lambda_2)(\delta + \lambda_2)}\right)\right). \quad (40)$$

If $\inf_\lambda \{\lambda \in \sigma(\widehat{R}_{\lambda_1})\} = \inf_\lambda \{\lambda \in \sigma(R_{\lambda_1})\} = 0$, i.e. $\delta' = \delta = 0$, (40) is just $\mathcal{O}_p(\frac{1}{\lambda_2^2 \sqrt{n}}) = \mathcal{O}_p(n^{2\beta - \frac{1}{2}})$. The asymptotic convergence behavior depends on the growth rate of $\lambda_2 = n^{-\beta}$. To make the expectation of the difference between $\widehat{q}_a^*$ and $P_n q_a^*$ in RKHS norm converges to 0 as n goes to infinity, we need

$$2\beta - \frac{1}{2} < 0 \implies 0 < \beta < \frac{1}{4}.$$

If $\inf_\lambda \{\lambda \in \sigma(\widehat{R}_{\lambda_1})\} > 0$ and $\inf_\lambda \{\lambda \in \sigma(R_{\lambda_1})\} > 0$, i.e. $\delta' > 0, \delta > 0$, which are possible for compact operators on finite dimensional spaces, (40) is upper bounded by $\mathcal{O}_p(\frac{1}{\sqrt{n}}(\frac{1}{\delta} + \frac{1}{\delta'\delta})) = \mathcal{O}_p(n^{-\frac{1}{2}})$, which is irrelevant to the decay rate of λ_2 .

□

Corollary 2.1 (Second moment convergence)

Suppose kernel functions $k(\cdot, \cdot)$ and $l(\cdot, \cdot)$ are bounded and smooth. Given regularization parameter λ_1 which is a fixed positive number and $\lambda_2 = n^{-\beta}$ for $\beta > 0$, the second moment of the difference between empirical regularized kernel embedded solution $\widehat{q}_a^*$ and regularized kernel embedded solution q_a^* in RKHS norm converges to 0 as n goes to ∞ with convergence rate

$$E\|\widehat{q}_a^* - q_a^*\|_{F_n}^2 \leq \begin{cases} \mathcal{O}(n^{4\beta - 1}), & \text{if } \inf_\lambda \{\lambda \in \sigma(\widehat{R}_{\lambda_1})\} = \inf_\lambda \{\lambda \in \sigma(R_{\lambda_1})\} = 0, \\ \mathcal{O}(n^{-1}), & \text{if } \inf_\lambda \{\lambda \in \sigma(\widehat{R}_{\lambda_1})\} > 0 \text{ and } \inf_\lambda \{\lambda \in \sigma(R_{\lambda_1})\} > 0. \end{cases}$$

Proof:

We continue to use the notations in the proof of Theorem 2.3, and consider only the convergence rate of $E\|\hat{q}_a^* - P_n q_a^*\|_{F_n}^2$. Then, by Hölder's inequality, the second moment of the difference in RKHS norm can be upper bounded by

$$E\|\hat{q}_a^* - P_n q_a^*\|_{F_n}^2 \leq E[T_1^2] + E[T_2^2] + E[T_1 T_2] \leq E[T_1^2] + E[T_2^2] + \sqrt{E[T_1^2]E[T_2^2]}. \quad (41)$$

where

$$\begin{aligned} T_1 &:= \|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)} - (R_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)}\|_{F_n} \\ T_2 &:= \|(R_{\lambda_1} + \lambda_2 I)^{-1} \hat{C}_{(WX)(ZAX)} \hat{C}_{\lambda_1}^{-1} \hat{\mu}_{(WX)} - (R_{\lambda_1} + \lambda_2 I)^{-1} C_{(WX)(ZAX)} C_{\lambda_1}^{-1} \mu_{(WX)}\|_{F_n}. \end{aligned}$$

By (34), combined with the upper bounds of operator norms and empirical embedding norms, $E[T_1^2]$ can be upper bounded by

$$\begin{aligned} E[T_1^2] &\leq E \left[\|\hat{C}_{(WX)(ZAX)}\|_{\mathcal{L}(H_n, F_n)}^2 \|\hat{\mu}_{(WX)}\|_{F_n}^2 \|\hat{C}_{\lambda_1}^{-1}\|_{\mathcal{L}(H_n)}^2 \|(\hat{R}_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F_n)}^2 \right. \\ &\quad \left. \cdot \|(R_{\lambda_1} + \lambda_2 I)^{-1}\|_{\mathcal{L}(F_n)}^2 \|R_{\lambda_1} - \hat{R}_{\lambda_1}\|_{\mathcal{L}(F_n)}^2 \right] \\ &\leq \frac{t_\phi^4 t_\psi^2}{\lambda_1^2 (\delta' + \lambda_2)^2 (\delta + \lambda_2)^2} E \|R_{\lambda_1} - \hat{R}_{\lambda_1}\|_{\mathcal{L}(F_n)}^2. \end{aligned}$$

By (38) and $\sqrt{n}$ -consistency of (cross-)covariance embeddings, $E\|R_{\lambda_1} - \hat{R}_{\lambda_1}\|_{\mathcal{L}(F_n)}^2$ has an upper bound

$$\begin{aligned} E\|R_{\lambda_1} - \hat{R}_{\lambda_1}\|_{\mathcal{L}(F_n)}^2 &\leq E \left[\left(\frac{t_\phi t_\psi}{\lambda_1} \|\hat{C}_{(ZAX)(WX)} - C_{(ZAX)(WX)}\|_{HS(F_n, H_n)} + \frac{t_\phi t_\psi}{\lambda_1} \|\hat{C}_{(WX)(ZAX)} - C_{(WX)(ZAX)}\|_{HS(H_n, F_n)} \right. \right. \\ &\quad \left. \left. + \frac{t_\phi^2 t_\psi^2}{\lambda_1^2} \|\hat{C}_{WX} - C_{WX}\|_{HS(H_n)} \right)^2 \right], \end{aligned}$$

which has a convergence rate of $\mathcal{O}(\frac{1}{n})$ after expanded by Hölder's inequality. Hence, we have that the convergence rate of $E[T_1^2]$ is $\mathcal{O}(\frac{1}{\lambda_1^4 n})$.

By (39), $\sqrt{n}$ -consistency of mean and (cross-)covariance embeddings, we have the upper bound of $E[T_2^2]$, which is given by

$$\begin{aligned} E[T_2^2] &\leq E \left[\left(\frac{t_\phi}{\lambda_1 (\delta + \lambda_2)} \|\hat{C}_{(WX)(ZAX)} - C_{(WX)(ZAX)}\|_{HS(H_n, F_n)} + \frac{t_\phi^2 t_\psi}{\lambda_1^2 (\delta + \lambda_2)} \|(\hat{C}_{(WX)} - C_{(WX)})\|_{HS(H_n, F_n)} \right. \right. \\ &\quad \left. \left. + \frac{t_\phi t_\psi}{\lambda_1 (\delta + \lambda_2)} \|\hat{\mu}_{(WX)} - \mu_{(WX)}\|_{F_n} \right)^2 \right] \\ &= \mathcal{O}\left(\frac{1}{(\delta + \lambda_2)^2 n}\right). \end{aligned}$$

combining the convergence rates of $E[T_1^2]$ and $E[T_2^2]$, by (41), we have that the second moment of the difference in RKHS norm has a convergence rate of

$$\mathcal{O}\left(\frac{1}{(\delta' + \lambda_2)^2 (\delta + \lambda_2)^2 n} + \frac{1}{(\delta + \lambda_2)^2 n} + \frac{1}{(\delta' + \lambda_2) (\delta + \lambda_2)^2 n}\right). \quad (42)$$

If $\inf_\lambda \{\lambda \in \sigma(\hat{R}_{\lambda_1})\} = \inf_\lambda \{\lambda \in \sigma(R_{\lambda_1})\} = 0$, i.e. $\delta' = \delta = 0$, (42) is just $\mathcal{O}(\frac{1}{\lambda_1^2 n}) = \mathcal{O}(n^{4\beta-1})$. When $\beta < \frac{1}{4}$, the second moment of the difference in RKHS norm converges to 0 as n goes to ∞ .

If $\inf_\lambda \{\lambda \in \sigma(\hat{R}_{\lambda_1})\} > 0$ and $\inf_\lambda \{\lambda \in \sigma(R_{\lambda_1})\} > 0$, i.e. $\delta' > 0, \delta > 0$, (42) is upper bounded by $\mathcal{O}_p(\frac{1}{\delta'^2 \delta^2 n} + \frac{1}{\delta^2 n} + \frac{1}{\delta' \delta^2 n}) = \mathcal{O}_p(n^{-1})$.

□

2.5 Kernel embedded doubly robust estimator, consistency and asymptotic normality

The kernel embedded ATE estimator built on doubly robust standardization formula (3) is given by

$$\begin{aligned}
\hat{\chi} &= \frac{1}{n} \sum_{i=1}^n [\mathbb{1}_{a_i=1} \hat{q}_1^*(z_i, 1, x_i) [y_i - \hat{h}_1^*(w_i, 1, x_i)] - \mathbb{1}_{a_i=0} \hat{q}_0^*(z_i, 0, x_i) [y_i - \hat{h}_0^*(w_i, 0, x_i)] + \hat{h}_1^*(w_i, 1, x_i) - \hat{h}_0^*(w_i, 0, x_i)] \\
&= \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=1} \hat{q}_1^*(z_i, 1, x_i) [y_i - \hat{h}_1^*(w_i, 1, x_i)] + \hat{h}_1^*(w_i, 1, x_i) \right\} - \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=0} \hat{q}_0^*(z_i, 0, x_i) [y_i - \hat{h}_0^*(w_i, 0, x_i)] + \hat{h}_0^*(w_i, 0, x_i) \right\} \\
&= \hat{E}[Y^1] - \hat{E}[Y^0].
\end{aligned} \tag{43}$$

We claim that under some regularization conditions, our kernel embedded ATE estimator is consistent to the true ATE and asymptotically normal when sample size $n \rightarrow \infty$.

2.5.1 Consistency

Theorem 2.4 (Consistency of the kernel embedded ATE estimator)

Assume the following conditions are true:

1. $\text{Var}(\mathbb{1}_{A=a} q_a^\dagger(Z, a, X)(Y - h_a^\dagger(W, a, X)) + h_a^\dagger(W, a, X)) < \infty$;
2. $E_{ZAX}[(E_Y[Y|Z, A, X])^2] < \infty$;
3. Kernel functions are bounded and smooth. Specifically, $\|\psi(\cdot)\|_F \leq t_\psi$, $\|\pi(\cdot)\|_G \leq t_\pi$ and $\|h_a^\dagger\|_G \leq t_h$.
4. The dataset used to estimate the ATE is independent of the dataset constructing the bridge function estimators.

Suppose regularizer $\lambda_2 = n^{-\beta}$, $\beta > 0$, with spectra of $\hat{R}_{\lambda_1}$ and R_{λ_1} bounded away from 0 from below. Then if the assumption 2.1 holds, under the distribution family satisfying the assumption 1.1 and 1.2, the kernel embedded doubly robust ATE estimator (43) is a consistent estimator of the true ATE given by doubly robust standardization formula (3), i.e.

$$\lim_{n \rightarrow \infty} E|\hat{\chi} - \chi| = 0.$$

The rate of convergence is at most $\begin{cases} \mathcal{O}_p(n^{-\beta}), 0 < \beta < \frac{1}{2}, \\ \mathcal{O}_p(n^{-\frac{1}{2}}), \beta \geq \frac{1}{2}. \end{cases}$

Proof:

Define

$$\begin{aligned}
\chi^\dagger &= E[\mathbb{1}_{A=1} q_1^\dagger(Z, 1, X)[Y - h_1^\dagger(W, 1, X)] - \mathbb{1}_{A=0} q_0^\dagger(Z, 0, X)[Y - h_0^\dagger(W, 0, X)] + h_1^\dagger(W, 1, X) - h_0^\dagger(W, 0, X)] \\
&= E^\dagger[Y^1] - E^\dagger[Y^0].
\end{aligned}$$

We first show $\lim_{n \rightarrow \infty} E|\hat{\chi} - \chi^\dagger| = 0$ by proving $\lim_{n \rightarrow \infty} E|\hat{E}[Y^a] - E^\dagger[Y^a]| = 0$. Then state that the definition of universal kernel guarantees $\chi^\dagger$ to approximate χ arbitrarily well. In the proof we will use the triangle inequality, Cauchy-Schwartz inequality, Hölder's inequality, Jensen's inequality and the reproducing property for reproducing kernel functions without mentioning.

$$\begin{aligned}
& E \left| \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} \widehat{q}_a^*(z_i, a, x_i) (y_i - \widehat{h}_a(w_i, a, x_i)) + \widehat{h}_a(w_i, a, x_i) \right\} - E \left[\mathbb{1}_{A=a} q_a^\dagger(Z, a, X) (Y - h_a^\dagger(W, a, X)) + h_a^\dagger(W, a, X) \right] \right| \\
& \leq E \left| \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} \widehat{q}_a^*(z_i, a, x_i) (y_i - \widehat{h}_a(w_i, a, x_i)) + \widehat{h}_a(w_i, a, x_i) \right\} - \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} q_a^\dagger(z_i, a, x_i) (y_i - h_a^\dagger(w_i, a, x_i)) + h_a^\dagger(w_i, a, x_i) \right\} \right| \\
& \quad + E \left| \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} q_a^\dagger(z_i, a, x_i) (y_i - h_a^\dagger(w_i, a, x_i)) + h_a^\dagger(w_i, a, x_i) \right\} - E \left[\mathbb{1}_{A=a} q_a^\dagger(Z, a, X) (Y - h_a^\dagger(W, a, X)) + h_a^\dagger(W, a, X) \right] \right| \\
& := l_1 + l_2
\end{aligned}$$

Part i) Bounding l_1 :

$$\begin{aligned}
& E \left| \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} \widehat{q}_a^*(z_i, a, x_i) (y_i - \widehat{h}_a(w_i, a, x_i)) + \widehat{h}_a(w_i, a, x_i) \right\} - \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} q_a^\dagger(z_i, a, x_i) (y_i - h_a^\dagger(w_i, a, x_i)) + h_a^\dagger(w_i, a, x_i) \right\} \right| \\
& \leq \frac{1}{n} \sum_{i=1}^n E \left| \mathbb{1}_{a_i=a} \widehat{q}_a^*(z_i, a, x_i) (y_i - \widehat{h}_a(w_i, a, x_i)) + \widehat{h}_a(w_i, a, x_i) - \mathbb{1}_{a_i=a} q_a^\dagger(z_i, a, x_i) (y_i - h_a^\dagger(w_i, a, x_i)) - h_a^\dagger(w_i, a, x_i) \right| \\
& = E \left| \mathbb{1}_{A=a} \widehat{q}_a^*(Z, a, X) [Y - \widehat{h}_a(W, a, X)] + \widehat{h}_a(W, a, X) - \mathbb{1}_{A=a} q_a^\dagger(Z, a, X) (Y - h_a^\dagger(W, a, X)) - h_a^\dagger(W, a, X) \right| \\
& \leq E \left| \mathbb{1}_{A=a} Y (\widehat{q}_a^*(Z, a, X) - q_a^\dagger(Z, a, X)) \right| + E \left| \widehat{h}_a(W, a, X) - h_a^\dagger(W, a, X) \right| \\
& \quad + E \left| \mathbb{1}_{A=a} \widehat{q}_a^*(Z, a, X) \widehat{h}_a(W, a, X) - \mathbb{1}_{A=a} q_a^\dagger(Z, a, X) h_a^\dagger(W, a, X) \right| \\
& := l_{11} + l_{12} + l_{13}.
\end{aligned}$$

For l_{11} ,

$$\begin{aligned}
l_{11} &:= E \left| \mathbb{1}_{A=a} Y (\widehat{q}_a^*(Z, a, X) - q_a^\dagger(Z, a, X)) \right| \\
&= E_{ZAX} [|\mathbb{1}_{A=a} (\widehat{q}_a^*(Z, a, X) - q_a^\dagger(Z, a, X))| E[Y|Z, A, X]] \\
&= E_{ZAX} [\langle \widehat{q}_a^* - q_a^\dagger, \psi(Z, A, X) \rangle_F E[Y|Z, A, X]] \\
&\leq E_{ZAX} [\|\widehat{q}_a^* - q_a^\dagger\|_F \|\psi(Z, A, X)\|_F E[Y|Z, A, X]] \\
&\leq (E\|\widehat{q}_a^* - q_a^\dagger\|_F^2)^{\frac{1}{2}} (E_{ZAX} [(E[Y|Z, A, X])^2])^{\frac{1}{2}} t_\psi.
\end{aligned} \tag{44}$$

To bound (44), we need the upper bound of $E\|\widehat{q}_a^* - q_a^\dagger\|_F^2$.

$$\begin{aligned}
E\|\widehat{q}_a^* - q_a^\dagger\|_F^2 &= E\|\widehat{q}_a^* - q_a^* + q_a^* - q_a^\dagger\|_F^2 \\
&= E\|\widehat{q}_a^* - q_a^*\|_F^2 + E\|q_a^* - q_a^\dagger\|_F^2 + 2E \left[\langle \widehat{q}_a^* - q_a^*, q_a^* - q_a^\dagger \rangle_F \right] \\
&\leq E\|\widehat{q}_a^* - q_a^*\|_F^2 + E\|q_a^* - q_a^\dagger\|_F^2 + 2E \left[\|\widehat{q}_a^* - q_a^*\|_F \|q_a^* - q_a^\dagger\|_F \right] \\
&\leq E\|\widehat{q}_a^* - q_a^*\|_F^2 + E\|q_a^* - q_a^\dagger\|_F^2 + 2(E\|\widehat{q}_a^* - q_a^*\|_F^2)^{\frac{1}{2}} (E\|q_a^* - q_a^\dagger\|_F^2)^{\frac{1}{2}}.
\end{aligned}$$

By Theorem 2.2 and Corollary 2.1, and Remark 2.2, the upper bound of the rate of convergence of right hand side is

$$\mathcal{O}(n^{-1}) + \mathcal{O}(n^{-2\beta}) + \mathcal{O}(n^{-\beta-\frac{1}{2}}). \tag{45}$$

With the same inequality technique, the analysis of l_{12} produces the rate of convergence that is exactly the same as (45). Then, for both l_{11} and l_{12} , the asymptotic convergence happens when $\begin{cases} -2\beta < 0, \\ -\beta - \frac{1}{2} < 0 \end{cases}$, with an

intersection $\beta > 0$. The dominant rate of convergence is given by $\begin{cases} \mathcal{O}(n^{-\beta}), 0 < \beta < \frac{1}{2}, \\ \mathcal{O}(n^{-\frac{1}{2}}), \beta \geq \frac{1}{2}. \end{cases}$

For l_{13} ,

$$\begin{aligned} l_{13} &:= E \left| \mathbb{1}_{A=a} \widehat{q}_a^*(Z, a, X) \widehat{h}_a(W, a, X) - \mathbb{1}_{A=a} q_a^\dagger(Z, a, X) h_a^\dagger(W, a, X) \right| \\ &\leq E \left| \mathbb{1}_{A=a} \widehat{q}_a^*(Z, a, X) (\widehat{h}_a(W, a, X) - h_a^\dagger(W, a, X)) \right| + E \left| \mathbb{1}_{A=a} (\widehat{q}_a^*(Z, a, X) - q_a^\dagger(Z, a, X)) h_a^\dagger(W, a, X) \right|. \end{aligned} \quad (46)$$

We only derive the upper bound of the rate of convergence of the second term on the right hand side, since the two terms share the same structure.

$$\begin{aligned} E \left| \mathbb{1}_{A=a} (\widehat{q}_a^*(Z, a, X) - q_a^\dagger(Z, a, X)) h_a^\dagger(W, A, X) \right| &= E \left| \left\langle h_a^\dagger, \pi(W, a, X) \right\rangle_G \left\langle \widehat{q}_a^* - q_a^\dagger, \psi(Z, A, X) \right\rangle_F \right| \\ &\leq E [\|h_a^\dagger\|_G \|\pi(W, A, X)\|_G \|\widehat{q}_a^* - q_a^\dagger\|_F \|\psi(Z, A, X)\|_F] \\ &\leq t_h t_\pi t_\psi E \|\widehat{q}_a^* - q_a^\dagger\|_F \\ &\leq t_h t_\pi t_\psi (E \|\widehat{q}_a^* - q_a^\dagger\|_F + E \|q_a^* - q_a^\dagger\|_F). \end{aligned}$$

By Theorem 2.2 and 2.3, and Remark 2.2, the upper bound of rate of convergence of right hand side is

$$\mathcal{O}(n^{-\frac{1}{2}}) + \mathcal{O}(n^{-\beta}).$$

The asymptotic convergence happens when $\beta > 0$. The dominant rate of convergence is given by $\begin{cases} \mathcal{O}(n^{-\beta}), 0 < \beta < \frac{1}{2}, \\ \mathcal{O}(n^{-\frac{1}{2}}), \beta \geq \frac{1}{2}. \end{cases}$

Part ii) Bounding l_2 : By the assumption of the finite variance of $\mathbb{1}_{A=a} q_a^\dagger(Z, a, X)(Y - h_a^\dagger(W, a, X)) + h_a^\dagger(W, a, X)$, l_2 can be upper bounded by

$$\begin{aligned} l_2 &:= E \left| \frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} q_a^\dagger(z_i, a, x_i) (y_i - h_a^\dagger(w_i, a, x_i)) + h_a^\dagger(w_i, a, x_i) \right\} - E \left[\mathbb{1}_{A=a} q_a^\dagger(Z, a, X) (Y - h_a^\dagger(W, a, X)) + h_a^\dagger(W, a, X) \right] \right| \\ &\leq \sqrt{\text{Var} \left(\frac{1}{n} \sum_{i=1}^n \left\{ \mathbb{1}_{a_i=a} q_a^\dagger(z_i, a, x_i) (y_i - h_a^\dagger(w_i, a, x_i)) + h_a^\dagger(w_i, a, x_i) \right\} \right)} \\ &= \frac{1}{\sqrt{n}} \sqrt{\text{Var} \left(\mathbb{1}_{A=a} q_a^\dagger(Z, a, X) (Y - h_a^\dagger(W, a, X)) + h_a^\dagger(W, a, X) \right)} \\ &= \mathcal{O}(n^{-\frac{1}{2}}). \end{aligned}$$

Part iii) Bounding $E|\widehat{\chi} - \chi|$:

Since $E|\widehat{\chi} - \chi^\dagger| \leq E|\widehat{E}[Y^1] - E^\dagger[Y^1]| + E|\widehat{E}[Y^0] - E^\dagger[Y^0]|$, combining the asymptotic behavior of l_1 and l_2 , we have the rate of convergence of $E|\widehat{\chi} - \chi^\dagger|$ by $\begin{cases} \mathcal{O}(n^{-\beta}), 0 < \beta < \frac{1}{2}, \\ \mathcal{O}(n^{-\frac{1}{2}}), \beta \geq \frac{1}{2}. \end{cases}$ By the definition of universal kernel, $\chi^\dagger$ should approximate the true ATE χ arbitrarily well. Hence, $\widehat{\chi}$ is a consistent estimator of χ . $\square$

2.5.2 Asymptotic normality

We discuss the asymptotic normality of the PDR estimator (43) by the double machine learning (DML) method proposed by Chernozhukov et al. [3]. Based on the proof using a simplified assumption given by Syrgkanis in [14], we show that, even though the Neyman orthogonality is satisfied, the PDR estimator built with dual kernel embedding method is asymptotically normal only when λ_1 is an effective stabilizer and the spectra of $\widehat{R}_{\lambda_1}$ and R_{λ_1} are bounded away from 0 from below.

The DML method is proposed to give an asymptotically normal estimator of a parameter lying in a finite dimensional space while a nuisance parameter exists. One of the situations discussed that is applicable with

DML is to estimate the parameter χ from moment condition $E[m(\mathcal{O}; \chi, \eta)] = 0$, where m is a known function in $L_2(P_{\mathcal{O}})$, $\mathcal{O}$ is all the observed random variables, and η is the nuisance parameter. Suppose χ_0 and η_0 are the true parameters. The standard procedure of DML is first to construct a Neyman orthogonal function m satisfying $E[\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta_0)(\hat{\eta} - \eta_0)] = 0$, meaning that the function is not sensitive to the first order perturbation of the estimator $\hat{\eta}$. Then use cross-fitting to split the training data and estimate the nuisance parameter on an auxiliary dataset to make sure the Neyman orthogonality works. If the estimator $\hat{\eta}$ has a decaying rate of $\mathcal{O}_p(n^{-\frac{1}{4}})$ to η_0 , then the estimator $\hat{\chi}$ built with $\hat{\eta}$ and the rest dataset is asymptotically normal with a covariance matrix Σ , i.e. $\sqrt{n}(\hat{\chi} - \chi_0) \rightarrow \mathcal{N}(0, \Sigma)$.

As for estimating ATE from PDR estimator, we write the moment condition by

$$E[\mathbb{1}_{A=1}q_1(Z, 1, X)[Y - h_1(W, 1, X)] - \mathbb{1}_{A=0}q_0(Z, 0, X)[Y - h_0(W, 0, X)] + h_1(W, 1, X) - h_0(W, 0, X) - \chi] = 0. \quad (47)$$

It is clear that in the proximal causal inference, the nuisance parameter η consists of the four bridge functions (q_1, q_0, h_1, h_0) .

We first introduce the simplified Neyman orthogonality and show that this requirement is satisfied.

Definition 2.1 (Simplified Neyman Orthogonality)

The function m is orthogonal at point (χ_0, η_0) if it satisfies

$$E[\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta_0)^T (\hat{\eta} - \eta_0)] = 0,$$

for any $\hat{\eta}$ that is an estimator of η built from some machine learning method.

As the Neyman orthogonality is unnecessary to discuss in our case, we suggest readers to check Definition 2.1 in [3] for a formal treatment.

Lemma 2.5 The function in moment condition (47) is orthogonal at the point with true parameter $(\chi_0, \eta^\dagger)$ ³ with respect to all the nuisance parameter estimator $\hat{\eta}$.

Proof:

$$\begin{aligned} E[\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger)] &= E \left[\begin{bmatrix} \nabla_{q_1} m(\mathcal{O}; \chi_0, \eta^\dagger) \\ \nabla_{q_0} m(\mathcal{O}; \chi_0, \eta^\dagger) \\ \nabla_{h_1} m(\mathcal{O}; \chi_0, \eta^\dagger) \\ \nabla_{h_0} m(\mathcal{O}; \chi_0, \eta^\dagger) \end{bmatrix}^T (\hat{\eta} - \eta^\dagger) \right] = E \left[\begin{bmatrix} \mathbb{1}_{A=1}(Y - h_1^\dagger(W, 1, X)) \\ -\mathbb{1}_{A=0}(Y - h_0^\dagger(W, 0, X)) \\ -\mathbb{1}_{A=1}q_1^\dagger(Z, 1, X) + 1 \\ \mathbb{1}_{A=0}q_0^\dagger(Z, 0, X) - 1 \end{bmatrix}^T (\hat{\eta} - \eta^\dagger) \right] \\ &= E \left[\begin{bmatrix} \mathbb{1}_{A=1}(Y - h_1^\dagger(W, 1, X)) \\ -\mathbb{1}_{A=0}(Y - h_0^\dagger(W, 0, X)) \\ -\mathbb{1}_{A=1}q_1^\dagger(Z, 1, X) + 1 \\ \mathbb{1}_{A=0}q_0^\dagger(Z, 0, X) - 1 \end{bmatrix}^T \begin{bmatrix} \hat{q}_1(Z, 1, X) - q_1^\dagger(Z, 1, X) \\ \hat{q}_0(Z, 1, X) - q_0^\dagger(Z, 1, X) \\ \hat{h}_1^*(W, 1, X) - h_1^\dagger(W, 1, X) \\ \hat{h}_0^*(W, 1, X) - h_0^\dagger(W, 1, X) \end{bmatrix} \right] \\ &= E_{ZX} [(\hat{q}_1(Z, 1, X) - q_1^\dagger(Z, 1, X))E[\mathbb{1}_{A=1}(Y - h_1^\dagger(W, 1, X))|Z, X]] \\ &\quad + E_{ZX} [(\hat{q}_1(Z, 1, X) - q_1^\dagger(Z, 1, X))E[-\mathbb{1}_{A=0}(Y - h_0^\dagger(W, 0, X))|Z, X]] \\ &\quad + E_{WX} [(\hat{h}_1^*(W, 1, X) - h_1^\dagger(W, 1, X))E[-\mathbb{1}_{A=1}q_1^\dagger(Z, 1, X) + 1|W, X]] \\ &\quad + E_{WX} [(\hat{h}_1^*(W, 1, X) - h_1^\dagger(W, 1, X))E[\mathbb{1}_{A=0}q_0^\dagger(Z, 0, X) - 1|W, X]] \end{aligned}$$

By the integral equation determining the existence of the bridge functions, we immediately have that the above moment function is 0. □

³Since $\eta^\dagger$ approximates η^* arbitrarily well, we deem $\eta^\dagger$ solves the integral equations.

Next we show that the rate of convergence of estimators $\widehat{q}^*, \widehat{h}^*$ built by dual kernel embedding method to the true boundedly continuous bridge functions is able to reach $\mathcal{O}_p(n^{-\frac{1}{4}})$ when the singular values of $\widehat{R}_{\lambda_1}$ and R_{λ_1} are bounded away from 0 from below.

Lemma 2.6 *Suppose the spectra of $\widehat{R}_{\lambda_1}$ and R_{λ_1} are bounded away from 0 from below. Let $\lambda_2 = n^{-\beta}$ for $\beta > 0$ to make the kernel embedded estimators $\widehat{q}_a^*, \widehat{h}_a^*$ asymptotically converges to the true bridge functions $q_a^\dagger, h_a^\dagger$. Then the rate of convergence of the square difference in RKHS norm is able to reach $\mathcal{O}_p(n^{-\frac{1}{2}})$ when $\beta > \frac{1}{4}$.*

Proof:

Recall that in Theorem 2.2 and Theorem 2.3 we have the asymptotic rate of convergences given by

$$\begin{aligned}\|\widehat{q}_a^* - q_a^\dagger\|_F^2 &\leq \|\widehat{q}_a^* - q_a^*\|_F^2 + \|q_a^* - q_a^\dagger\|_F^2 + 2\|\widehat{q}_a^* - q_a^*\|_F \|q_a^* - q_a^\dagger\|_F \\ &= \mathcal{O}_p(n^{-1}) + \mathcal{O}_p(n^{-2\beta}) + \mathcal{O}(n^{-\beta-\frac{1}{2}}).\end{aligned}$$

The three curves -1 , -2β and $-\beta - \frac{1}{2}$ come across at $\beta = \frac{1}{2}$. When $0 < \beta < \frac{1}{2}$, we have $-2\beta > -\beta - \frac{1}{2} > -1$, while $-1 > -\beta - \frac{1}{2} > -2\beta$ on the other side of the intersection. So, the dominant rate of convergence for the two cases are $\begin{cases} \mathcal{O}_p(n^{-2\beta}), & \text{if } 0 < \beta < \frac{1}{2}, \\ \mathcal{O}_p(n^{-1}), & \text{if } \beta \geq \frac{1}{2}. \end{cases}$ To make the rate of convergence faster than $\mathcal{O}_p(n^{-\frac{1}{2}})$, we need $\beta \in (\frac{1}{4}, \frac{1}{2}) \cup [\frac{1}{2}, +\infty)$, which is just $\beta > \frac{1}{4}$. □

However, when the spectra of $\widehat{R}_{\lambda_1}$ and R_{λ_1} are not bounded away from 0 from below, the rate of convergence may become slower than $\mathcal{O}_p(n^{-\frac{1}{4}})$ or doesn't exist considering the speed of $\|q_a^* - q_a^\dagger\|_F^2$. This is a fatal uncertainty that can't guarantee the asymptotic normality of the kernel embedded PDR estimator.

Theorem 2.5 (Asymptotic normality of kernel embedded PDR estimator of ATE)

When

1. $\text{Var}(\mathbb{1}_{A=1}q_1^\dagger(Z, 1, X)[Y - h_1^\dagger(W, 1, X)] - \mathbb{1}_{A=0}q_0^\dagger(Z, 0, X)[Y - h_0^\dagger(W, 0, X)] + h_1^\dagger(W, 1, X) - h_0^\dagger(W, 0, X)) < \infty$;
2. The spectra of $\widehat{R}_{\lambda_1}$ and R_{λ_1} are bounded away from 0 from below;
3. The kernel functions are bounded and smooth;
4. The regularizer $\lambda_2 = n^{-\beta}$ has an exponent larger than $\frac{1}{4}$,

the DML method succeeds to construct an asymptotically normal PDR estimator of ATE using the kernel embedded estimator of the two bridge functions, i.e.

$$\sqrt{n_c}(\widehat{\chi} - \chi_0) \xrightarrow{d} \mathcal{N}(0, \Sigma),$$

where n_c is the size of training set and

$$\Sigma = \text{Var}(\mathbb{1}_{A=1}q_1^\dagger(Z, 1, X)[Y - h_1^\dagger(W, 1, X)] - \mathbb{1}_{A=0}q_0^\dagger(Z, 0, X)[Y - h_0^\dagger(W, 0, X)] + h_1^\dagger(W, 1, X) - h_0^\dagger(W, 0, X))$$

Proof:

We split the dataset $\mathcal{D}$ into an auxiliary set $\mathcal{D}_I$ to perform the dual kernel embedding method to construct the kernel embedded estimators $\widehat{q}_a^*, \widehat{h}_a^*$, and a training set $\mathcal{D}_{I^c}$ to estimate the ATE with PDR estimator. Here $\mathcal{D}_{I^c} \cup \mathcal{D}_I = \mathcal{D}$ and $\mathcal{D}_{I^c} \cap \mathcal{D}_I = \emptyset$. We assume the sample sizes of the auxiliary set and training set are $n = |\mathcal{D}_I|$ and $n_c = |\mathcal{D}_{I^c}|$ with the same magnitude.

The PDR estimator $\hat{\chi}$ solves the empirical moment condition of (47) with $\hat{\eta} = (q_a^*, h_a^*)$, which means $\frac{1}{n_c} \sum_{i \in I^c} m(\mathcal{O}_i; \hat{\chi}, \hat{\eta}) = 0$. By mean value theorem, we have

$$\begin{aligned} \frac{1}{n_c} \sum_{i \in I^c} m(\mathcal{O}_i; \hat{\chi}, \hat{\eta}) &= \frac{1}{n_c} \sum_{i \in I^c} m(\mathcal{O}_i; \chi_0, \hat{\eta}) + \frac{1}{n_c} \sum_{i \in I^c} \nabla_{\chi} m(\mathcal{O}_i; \tilde{\chi}, \hat{\eta})(\hat{\chi} - \chi_0) \\ 0 &= \frac{1}{n_c} \sum_{i \in I^c} m(\mathcal{O}_i; \chi_0, \hat{\eta}) - (\hat{\chi} - \chi_0) \\ \implies \sqrt{n_c}(\hat{\chi} - \chi_0) &= \frac{1}{\sqrt{n_c}} \sum_{i \in I^c} m(\mathcal{O}_i; \chi_0, \hat{\eta}), \end{aligned} \quad (48)$$

where $\tilde{\chi} = \lambda \hat{\chi} + (1 - \lambda) \chi_0$, for $\lambda \in (0, 1)$. We Taylor expand the right hand side of (48) with respect to η to the order 2 around $\eta^\dagger = (q_a^\dagger, h_a^\dagger)$.

$$\begin{aligned} \frac{1}{\sqrt{n_c}} \sum_{i \in I^c} m(\mathcal{O}_i; \chi_0, \hat{\eta}) &= \underbrace{\frac{1}{\sqrt{n_c}} \sum_{i \in I^c} m(\mathcal{O}_i; \chi_0, \eta^\dagger)}_{I_1} + \underbrace{\frac{1}{\sqrt{n_c}} \sum_{i \in I^c} \nabla_{\eta} m(\mathcal{O}_i; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger)}_{I_2} \\ &\quad + \underbrace{\frac{1}{\sqrt{n_c}} \sum_{i \in I^c} (\hat{\eta} - \eta^\dagger)^T \nabla_{\eta\eta} m(\mathcal{O}_i; \chi_0, \eta^\dagger) (\hat{\eta} - \eta^\dagger)}_{I_3} + \frac{1}{\sqrt{n_c}} o(\|\hat{\eta} - \eta^\dagger\|^2). \end{aligned}$$

By the central limit theorem, $I_1 \rightarrow \mathcal{N}(0, \Sigma)$, with $\Sigma = E[m^2(\mathcal{O}; \chi_0, \eta^\dagger)]$, and $I_2 \rightarrow \mathcal{N}(E_{\mathcal{D}_{I^c}}[\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger)], \text{Var}_{\mathcal{D}_{I^c}}(\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger))$. Since Lemma 2.5 shows that⁴ $E_{\mathcal{D}_{I^c}}[\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger)] = 0$, we check the variance. By the bounded continuity assumption of the bridge functions, we assume $\|\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)\|_1^2$ is uniformly upper bounded by constant $B > 0$ over $\mathcal{O}$. In fact,

$$\begin{aligned} \text{Var}_{\mathcal{D}_{I^c}}(\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger)) &= E_{\mathcal{D}_{I^c}}[(\hat{\eta} - \eta^\dagger)^T \nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger) \nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger)] \\ &\leq E_{\mathcal{D}_{I^c}}[\|\hat{\eta} - \eta^\dagger\|_1^2 \|\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)\|_1^2] \\ &= B \sum_{a \in \{0, 1\}} \left[\|\hat{q}_a^* - q_a^\dagger\|^2 \|\tilde{\mu}_{(ZX)}\|^2 + \|\hat{h}_a^* - h_a^\dagger\|^2 \|\tilde{\mu}_{(WX)}\|^2 \right]. \end{aligned}$$

By the boundedness of RKHS elements, we have that $\text{Var}_{\mathcal{D}_{I^c}}(\nabla_{\eta} m(\mathcal{O}; \chi_0, \eta^\dagger)^T (\hat{\eta} - \eta^\dagger))$ converges to 0 in probability with a rate of convergence equivalent to the one of $\|\hat{q}_a^* - q_a^\dagger\|^2$. This means I_2 will asymptotically degenerate to 0 as $\|\hat{q}_a^* - q_a^\dagger\|^2$ and $\|\hat{h}_a^* - h_a^\dagger\|^2$ converge to 0.

As for I_3 , since $\nabla_{\eta\eta} m(\mathcal{O}_i; \chi_0, \eta^\dagger) = \begin{bmatrix} 0 & 0 & 1 - \mathbb{1}_{A=1} & -1 \\ 0 & 0 & 1 & \mathbb{1}_{A=0} - 1 \\ -\mathbb{1}_{A=1} & 0 & 0 & 0 \\ 0 & \mathbb{1}_{A=0} & 0 & 0 \end{bmatrix}$ is irrelevant to η and χ , we know

its eigenvalues are upper bounded by a positive constant which we denote by M for all $A \in \{0, 1\}$. Then as $n_c \rightarrow \infty$, $I_3 \lesssim M \sqrt{n_c} E_{\mathcal{D}_{I^c}}[\|\hat{\eta} - \eta^\dagger\|_1^2]$, which has a rate of convergence equivalent to the one of $\sqrt{n_c} \|\hat{q}_a^* - q_a^\dagger\|^2$.

Hence the rate of convergence of $I_2 + I_3$ is dominated by I_3 . As discussed in Lemma 2.6, we have that when $\beta > \frac{1}{4}$, the rate of convergence of $\|\hat{q}_a^* - q_a^\dagger\|^2$ is able to exceed $\mathcal{O}_p(n^{-\frac{1}{2}})$ and gives a convergence to 0 for I_3 .

Therefore, by (48), $\sqrt{n_c}(\hat{\chi} - \chi_0)$ asymptotically approximates to a normal distribution $\mathcal{N}(0, \Sigma)$, with

$$\begin{aligned} \Sigma &= E[m^2(\mathcal{O}; \chi_0, \eta^\dagger)] \\ &= \text{Var}(\mathbb{1}_{A=1} q_1^\dagger(Z, 1, X)[Y - h_1^\dagger(W, 1, X)] - \mathbb{1}_{A=0} q_0^\dagger(Z, 0, X)[Y - h_0^\dagger(W, 0, X)] + h_1^\dagger(W, 1, X) - h_0^\dagger(W, 0, X)), \end{aligned}$$

as $n, n_c \rightarrow \infty$.

□

With the same proof technique, we also have the asymptotic normality of the kernel embedded CERF $\hat{E}[Y^a]$.

⁴We denote $E[\cdot | \mathcal{D}_I]$ by $E_{\mathcal{D}_{I^c}}$ when both auxiliary and training sets appear.

Theorem 2.6 (Asymptotic normality of kernel embedded PDR estimator of CERF)

When

1. $\text{Var}(\mathbb{1}_{A=a} q_a^\dagger(Z, a, X)[Y - h_a^\dagger(W, a, X)] + h_a^\dagger(W, a, X)) < \infty$;
2. The spectra of $\widehat{R}_{\lambda_1}$ and R_{λ_1} are bounded away from 0 from below;
3. The kernel functions are bounded and smooth;
4. The regularizer $\lambda_2 = n_a^{-\beta}$ has an exponent larger than $\frac{1}{4}$,

the DML method succeeds to construct an asymptotically normal PDR estimator of CERF using the kernel embedded estimator of the two bridge functions, i.e.

$$\sqrt{n_{c_a}}(\widehat{E}[Y^a] - E[Y^a]) \xrightarrow{d} \mathcal{N}(0, \sigma_a^2),$$

where n_a is the size of auxiliary set, n_{c_a} is the size of training set and

$$\sigma_a^2 = \text{Var}(\mathbb{1}_{A=a} q_a^\dagger(Z, a, X)[Y - h_a^\dagger(W, a, X)] + h_a^\dagger(W, a, X)).$$

When applied to real world cases, the variance can be estimated by sample variance S of the kernel embedded PDR estimator $\widehat{\chi}$, since we can show S is consistent to the variance by the same trick used in Corollary 2.1 and Theorem 2.4. Then we can build up an asymptotic standard normal statistic with $\widehat{\chi}$ to perform hypothesis test. However, as stated in Remark 2.1, direct estimation of ATE with kernel embedded estimator $\widehat{\chi}$ may produce a solution that differs from the true value by a constant factor because of the uncertainty brought by the final least square solution. Hence, we estimate $E[Y^1]$ and $E[Y^0]$ respectively to approximate ATE with their difference, which is consistent to the true value when sample size is large. With the derived asymptotic normality and the sample variance of σ_a^2 , $a \in \{0, 1\}$, we are able to construct another asymptotic standard normal statistic with $\widehat{\chi} = \widehat{E}[Y^1] - \widehat{E}[Y^0]$ to conduct the hypothesis test on whether the true ATE χ_0 is 0.

Corollary 2.2 (Asymptotically standard normal statistic)

When $\widehat{E}[Y^1]$ and $\widehat{E}[Y^0]$ are asymptotically normal, whose variances $\widehat{\sigma}_1^2$ and $\widehat{\sigma}_0^2$ are consistent to σ_1^2 and σ_0^2 , we have

$$\frac{\widehat{\chi} - \chi_0}{\sqrt{\frac{\widehat{\sigma}_1^2}{n_{c1}} + \frac{\widehat{\sigma}_0^2}{n_{c0}}}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Proof:

As given by Theorem 2.6, we have

$$\sqrt{n_{c1}}(\widehat{E}[Y^1] - E[Y^1]) \xrightarrow{d} \mathcal{N}(0, \sigma_1^2), \quad \sqrt{n_{c0}}(\widehat{E}[Y^0] - E[Y^0]) \xrightarrow{d} \mathcal{N}(0, \sigma_0^2).$$

By the approach to constructing $\widehat{E}[Y^1]$ and $\widehat{E}[Y^0]$ given in Remark 2.1, $\widehat{E}[Y^1]$ is independent of $\widehat{E}[Y^0]$. Then as $\sqrt{n_{c1}}, \sqrt{n_{c2}} \rightarrow \infty$, we have

$$\widehat{\chi} = \widehat{E}[Y^1] - \widehat{E}[Y^0] \xrightarrow{d} \mathcal{N}(\chi_0, \frac{\sigma_1^2}{n_{c1}} + \frac{\sigma_0^2}{n_{c0}}).$$

By Slutsky's lemma, we further have

$$\frac{\widehat{\chi} - \chi_0}{\sqrt{\frac{\widehat{\sigma}_1^2}{n_{c1}} + \frac{\widehat{\sigma}_0^2}{n_{c0}}}} \xrightarrow{d} \mathcal{N}(0, 1).$$

□

With Corollary 2.2, we define the Wald ratio test for ATE in proximal causal inference with binary treatment $A = a \in \{0, 1\}$.

Definition 2.2 (Wald ratio test)

Given the asymptotic normal ATE estimator $\hat{\chi}$

$$\hat{\chi} = \hat{E}[Y^1] - \hat{E}[Y^0] \xrightarrow{d} \mathcal{N}(\chi_0, \frac{\sigma_1^2}{n_{c1}} + \frac{\sigma_0^2}{n_{c0}}),$$

considering the hypothesis test

$$H_0 : \chi_0 = 0 \leftrightarrow H_1 : \chi_0 \neq 0,$$

we define the Wald ratio statistic by

$$W := \frac{\hat{\chi}}{\sqrt{\frac{\hat{\sigma}_1^2}{n_{c1}} + \frac{\hat{\sigma}_0^2}{n_{c0}}}}.$$

which satisfies $W \xrightarrow{d} \mathcal{N}(0, 1)$, under the null hypothesis H_0 and when $\hat{\sigma}_1^2, \hat{\sigma}_0^2$ are consistent to σ_1^2, σ_0^2 . For a significance level α , let $z_{\frac{\alpha}{2}}$ be the upper $\frac{\alpha}{2}$ percentile of standard normal distribution, the rejection area of the hypothesis test is where $|W| \geq z_{\frac{\alpha}{2}}$. The confidence interval is

$$CI_\alpha := [\hat{\chi} - \sqrt{\frac{\hat{\sigma}_1^2}{n_{c1}} + \frac{\hat{\sigma}_0^2}{n_{c0}}} z_{\frac{\alpha}{2}}, \hat{\chi} + \sqrt{\frac{\hat{\sigma}_1^2}{n_{c1}} + \frac{\hat{\sigma}_0^2}{n_{c0}}} z_{\frac{\alpha}{2}}]$$

The p-value for this test is

$$P := 2Pr(|Z| \geq W), \text{ where } Z \sim \mathcal{N}(0, 1).$$

2.5.3 Numerical test on consistency and asymptotic normality

In this section, we explain the numerical test of the kernel embedded PDR estimator (43) and analyze its convergence behaviors on a synthetic dataset.

Data generation The data generation scheme of the numerical test is based on the graphical model 1. We assume the data is sequentially generated as follows.

$$\begin{aligned} U &= \epsilon_1 \\ X &= 0.5U + \epsilon_2 \\ Z &= 0.3U + 0.4X + \epsilon_3 \\ W &= 0.6U + 0.3X + \epsilon_4 \\ A &\sim \text{Bernoulli}(p), \quad p = E[\mathbf{1}_{0.3U+0.5X+0.7Z+\epsilon_5 > 0}] \\ Y^1 &= \mu_1 + 0.07U + 0.02X + 0.08W + 0.1\epsilon_6 \\ Y^0 &= \mu_0 + 0.03U + 0.04X + 0.03W + 0.1\epsilon_7 \\ Y &= \mathbf{1}_{A=1}Y^1 + \mathbf{1}_{A=0}Y^0, \end{aligned}$$

where $\epsilon_1, \dots, \epsilon_7$ are i.i.d. from $\mathcal{N}(0, 1)$.

According to the model, since $E[U] = E[X] = E[W] = 0$, the true ATE is given by

$$\chi = E[Y^1] - E[Y^0] = \mu_1 - \mu_0.$$

Kernel function To compute the empirical regularized kernel embedded solution $\hat{q}_a^*$ (19) by Proposition 2.3, we choose the Gaussian RBF kernel to construct the RKHSs. The kernel is given by

$$\begin{aligned} k((W, X), (W', X')) &:= \exp\{-\gamma\|[WX] - [W'X']\|^2\}, \\ \tilde{l}((Z, X), (Z', X')) &:= \exp\{-\gamma\|[ZX] - [Z'X']\|^2\}, \\ l((Z, A, X), (Z', A', X')) &:= \mathbf{1}_{A=a}\mathbf{1}_{A'=a} \exp\{-\gamma\|[ZX] - [Z'X']\|^2\}, \quad \forall a \in \{0, 1\}, \end{aligned}$$

where $[\cdot]$ is a concatenated vector, i.e. $\forall A, B \in \mathbb{R}^n$, then $[AB] = A \oplus B \in \mathbb{R}^{2n}$.

Hyperparameter selection Since matrix $L(K + n\lambda_1 I_n)^{-1}KL + n\lambda_2 L$ is not always invertible, we turn to the Tikhonov regularized solution of the kernel embedded coefficient $\hat{\Lambda} := (C^*C + \lambda I)C^*b$ (22). This leads to selection of 4 parameters: $\lambda \in (0, +\infty)$, $\gamma \in (0, +\infty)$, $\lambda_1 \in (0, +\infty)$ and $\beta \in (0, +\infty)$. As for λ and γ , we fix them by positive constants through out the hyperparameter tuning since they only influence the stability of the estimator but will not bring unexpected bias. Since in real world scenarios, the data distribution usually doesn't satisfy the lower bounded spectra of $\hat{R}_{\lambda_1}$ and R_{λ_1} , hence we can't expect a good convergence by merely setting a large enough β . To facilitate finding the optimal values, we choose some potentially optimal points and restrict the searching area of the parameters within

$$\begin{aligned}\lambda_1 &\in \{1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1\} \\ \beta &\in \{1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1, 1e+1, 1e+2\}.\end{aligned}$$

Recall that Proposition 2.1 shows that the dual function measures the error that the estimated bridge function brings to the integral equation (4). We use the empirical square error of the kernel embedded dual function

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{a_i=1} (\hat{u}_1^*(w_i, x_i))^2 + \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{a_i=0} (\hat{u}_0^*(w_i, x_i))^2 + \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{a_i=1} (\hat{u}_1^*(z_i, x_i))^2 + \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{a_i=0} (\hat{u}_0^*(z_i, x_i))^2 \quad (49)$$

as the loss function to choose hyperparameters.

Of note that for a set of observations, we split the dataset equally to build the kernel embedded estimators with one half, and estimate ATE using the other half. We shuffle the whole dataset for 3 times to generate 3 random splittings. The total loss is the average of the 3 losses (49) on all shuffled dataset.

When hyperparameter tuner returns the optimal values of λ_1 and β , we apply them directly to the following tests.

Test on consistency We conduct two simple experiments on the convergence of the ATE estimator with $\mu_1 = 2$ and $\mu_0 = -1$. For the first one, we want to test the mean absolute error (MAE) of the ATE estimator as sample size grows from 50 to 10050 with the interval of 500 between two adjacent points. We choose the hyperparameters from the potential ranges and apply the optimal set directly to the test. The second experiment is on the MAE as β grows when sample size is fixed to 3000, in which we choose the test points as $(1e-16, 1e-2, 1e-1, 1, 1e+1, 1e+2, 1e+3)$, and set $\lambda_1 = 1e-1$. Through out the two experiments, we fix $\lambda = 1e-1$ and $\gamma = 1e-3$, and run 100 times for each test point to compute average and standard deviation.

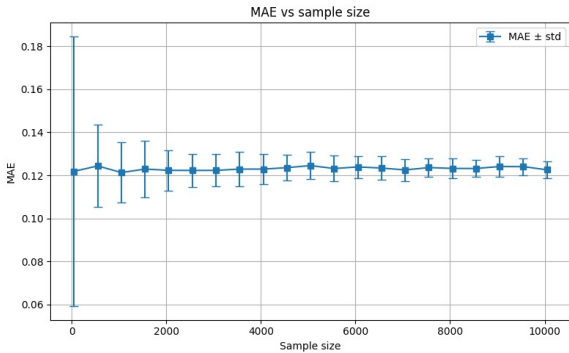


Figure 2: MAE vs. sample size

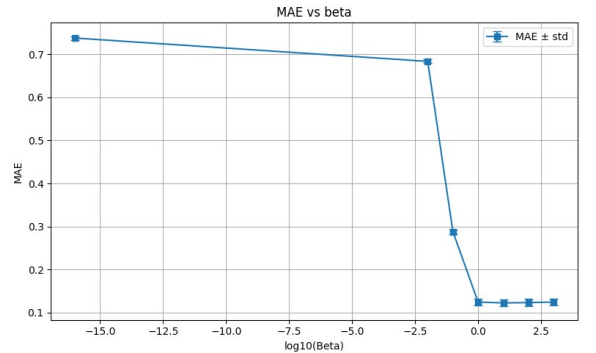


Figure 3: MAE vs. Beta

The results are shown in Figure 2 and 3.

The hyperparameter tuning process returns the optimal set of λ_1 and β by $(1, 100)$, which witnesses larger λ_1 will better increase the convergence speed of $\|\hat{R}_{\lambda_1} - R_{\lambda_1}\|_{\mathcal{L}(F)}$, as shown by (36), and therefore speed up the convergence of $\|\hat{q}_a^* - q_a^*\|_F$. In addition, this implies for the data distribution we are using, the spectra of operators $\hat{R}_{\lambda_1}$ and R_{λ_1} are bounded away from 0 from below because 0 is possibly not a part of the spectra any more for compact operators on finite dimensional spaces. This makes the optimal β as large as possible. In fact, Figure 3 helps illustrate this statement. Moreover, from Figure 2, we have that strong enough regularizer parameters have already maintained the consistency of the estimator when sample size is small but still can't control the variance. When sample size grows large, the variance of estimator gradually decreases.

Test on asymptotic normality We generate a dataset with sample size 10000 using the same synthetic data generation rule of the consistency test. The ATE estimation $\hat{\chi}$ on the dataset is 3.12. Then we perform 1000 bootstrap resamplings on the generated dataset, and calculate the estimated ATE for each bootstrapped dataset.

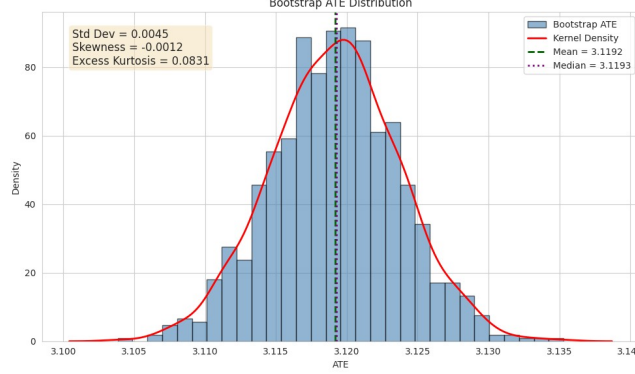


Figure 4: The histogram of the bootstrap ATE estimations

The histogram of the result, Figure 4, shows that the ATE estimations approximately form a normal distribution because the skewness and kurtosis coefficients are nearly 0.

Furthermore, we use centered bootstrap percentile test and compute p-value to compare with the p-value computed by the two-tailed test for the mean of the standard normal distribution. Denote by $(\tilde{\chi}_i)_{1 \leq i \leq 1000}$ the ATE estimations for all the bootstrap resampled datasets, and $\bar{\chi}$ the mean of them. The formula for p-value is given by

$$P_{bp} = \frac{1 + \sum_{i=1}^{1000} \mathbb{1}_{\tilde{\chi}_i - \bar{\chi} \geq |\hat{\chi}|}}{1 + 1000}.$$

In our test, $P_{bp} = 9.99e - 04 = \frac{1}{1001}$, meaning that $\hat{\chi}$ is outside the density formed by the centered bootstrap ATE estimations.

By sample deviation $S = 0.004472$ matching the standard deviation 0.0045 calculated in bootstrap experiment, the Z-statistic $\frac{\hat{\chi}}{\sqrt{S}} = \frac{3.12}{\sqrt{0.004472}}$ is 699.65, which locates at the extremely far right tail of the standard normal distribution. The p-value

$$P = 2 \times (1 - \Phi(699.65)) \approx 0,$$

matches P_{bp} observed by bootstrap test.

2.5.4 Sensitivity analysis

Considering a situation with invalid proxies, as shown by Figure 5, we discuss how will the symmetric direct influence $\zeta : Z \rightarrow Y, W \rightarrow A$ affects the model accuracy on estimating ATE.

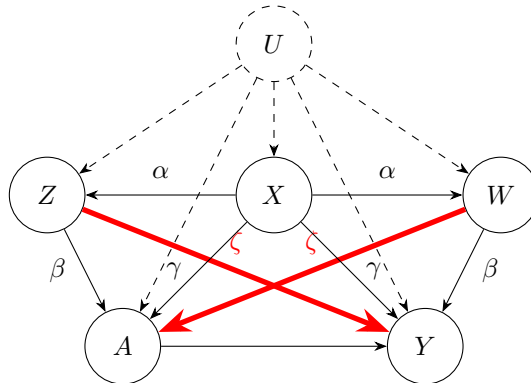


Figure 5: DAG of sensitivity analysis

We use the chance of the error of the first kind to measure the performance of the model in the sensitivity analysis. We define data generation schemes by negative (ATE=0) and positive (ATE $\neq$ 0) case generators, which are given in Table 1. For both generators, $\epsilon_1, \dots, \epsilon_7$ are i.i.d. from $\mathcal{N}(0,1)$.

Positive case	Negative case
$U = \epsilon_1$	$U = \epsilon_1$
$X = 0.05U + \epsilon_2$	$X = 0.05U + \epsilon_2$
$Z = 0.04U + \alpha X + \epsilon_3$	$Z = 0.04U + \alpha X + \epsilon_3$
$W = 0.04U + \alpha X + \epsilon_4$	$W = 0.04U + \alpha X + \epsilon_4$
$A \sim \text{Bernoulli}(p), p = E[\mathbb{1}_{0.03U + \gamma X + \beta Z + \zeta W + \epsilon_5 > 0}]$	$A \sim \text{Bernoulli}(p), p = E[\mathbb{1}_{0.03U + \gamma X + \beta Z + \zeta W + \epsilon_5 > 0}]$
$Y^1 \sim \text{Bernoulli}(p_1), p_1 = E[\mathbb{1}_{\mu_1 + 0.03U + \gamma X + \beta W + \zeta Z + \epsilon_6 > 0}]$	$Y^1 \sim \text{Bernoulli}(p_1), p_1 = E[\mathbb{1}_{\mu + 0.03U + \gamma X + \beta W + \zeta Z + \epsilon_6 > 0}]$
$Y^0 \sim \text{Bernoulli}(p_0), p_0 = E[\mathbb{1}_{\mu_0 + 0.03U + \gamma X + \beta W + \zeta Z + \epsilon_7 > 0}]$	$Y^0 \sim \text{Bernoulli}(p_0), p_0 = E[\mathbb{1}_{\mu + 0.03U + \gamma X + \beta W + \zeta Z + \epsilon_7 > 0}]$
$Y = \mathbb{1}_{A=1}Y^1 + \mathbb{1}_{A=0}Y^0$	$Y = \mathbb{1}_{A=1}Y^1 + \mathbb{1}_{A=0}Y^0$

Table 1: Data generation schemes of influence model in sensitivity analysis

We control the coefficients α , β and γ by 0.05 and test the value of ζ among 0, 0.05, 0.5, 2.

The first experiment tests the deviation of the p-value as ζ changes when the true ATE is 0. The result is given in Figure 6. It shows that as the common direct influence of the invalid proxies goes up, the observed p-value gradually becomes smaller than the expected p-value, meaning the increased chance of the error of the first kind of the model.

The second experiment tests the prediction accuracy of the model featured by curves of ROC and TP/FP vs. FDR. Here, for each test point of ζ , we set significance levels from 0.05 to 1 to maximally expand the range of FPR and FDR. Then, for each significance level, we generate ground truth by 500 positive ATEs and 500 negative ATEs with observations of equal sample size 1000 respectively. The model subsequently compute ATEs from the dataset and record all true/false positive and true/false negative ones. The result is shown in Figure 7. From the two curves, we have that as ζ rises to 2 which is non-negligible compared to the rest coefficients in the model, the prediction performance drops drastically. The ROC curve at $\zeta = 2$ indicates a comparatively low TPR for all FPR. Combined with the overall decreased TP number and the right shifted starting points of TP/FP vs. FDR curve, the result witnesses the increment on the error of the first kind when the direct influence of the proxies to the treatment and outcome that shouldn't have existed is large enough.

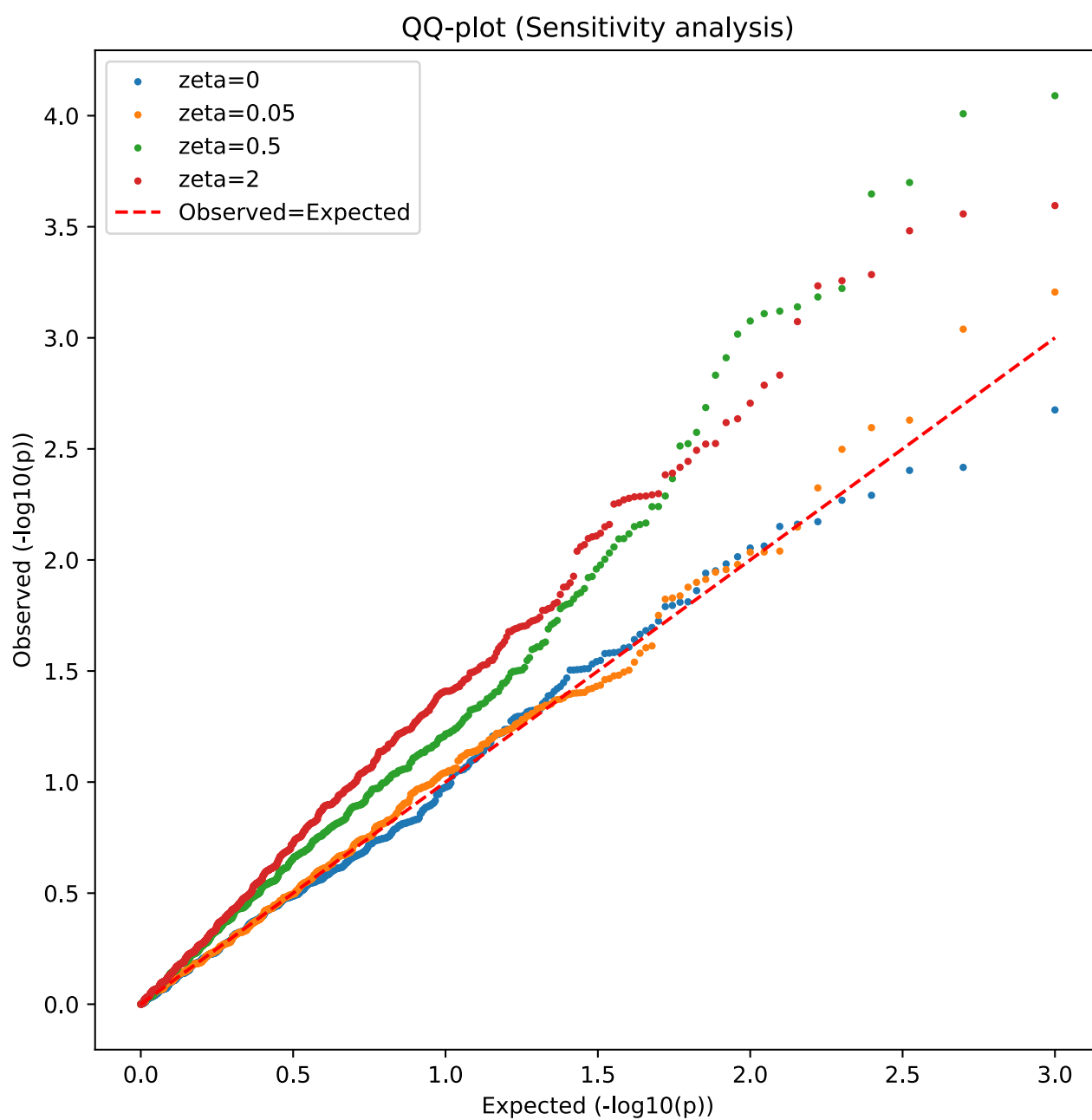


Figure 6: QQ-plot for the p-value when true ATE is 0

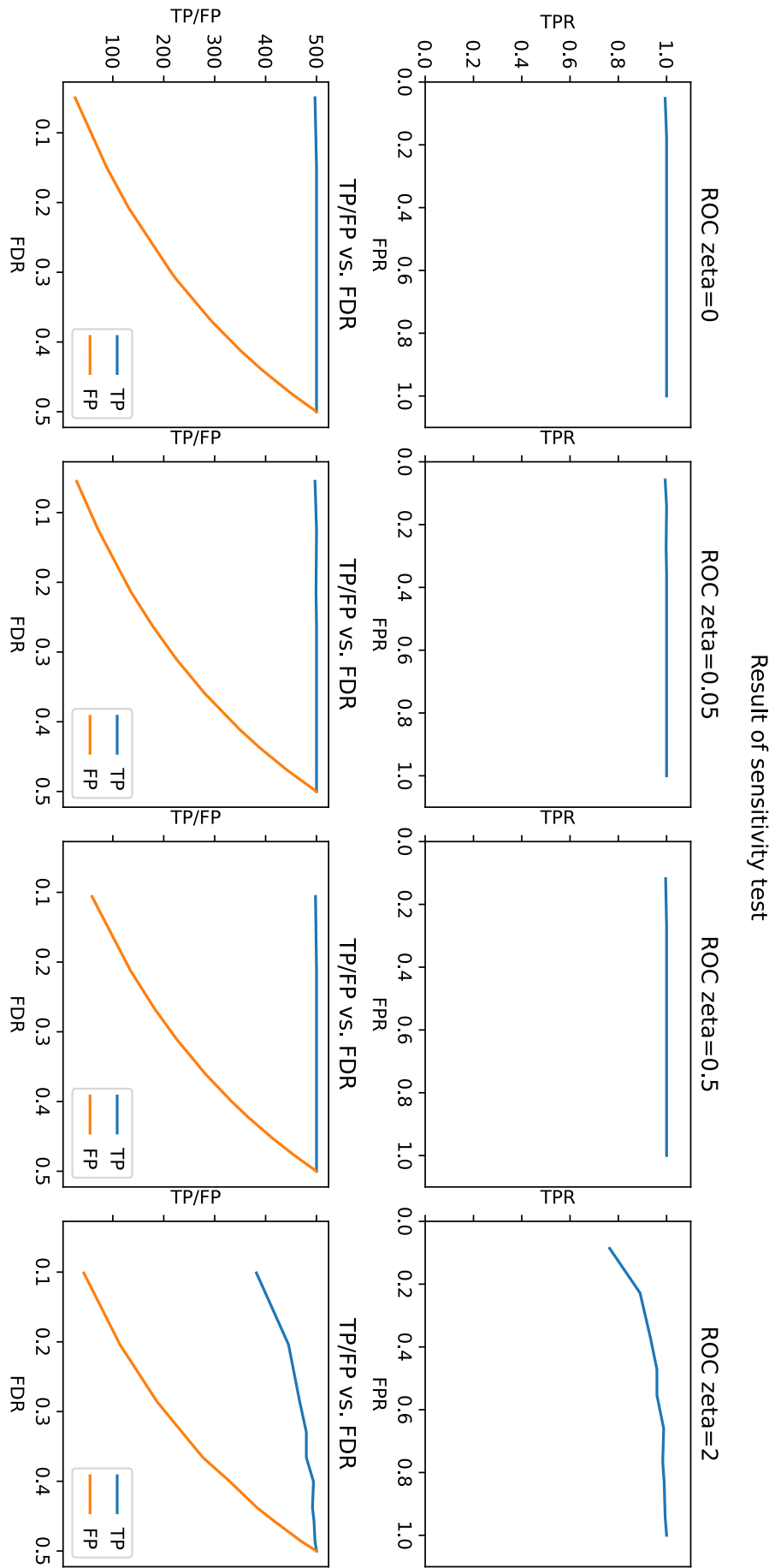


Figure 7: The overall prediction accuracy of the model when proxies are invalid

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